Configuration: Data Scenarios

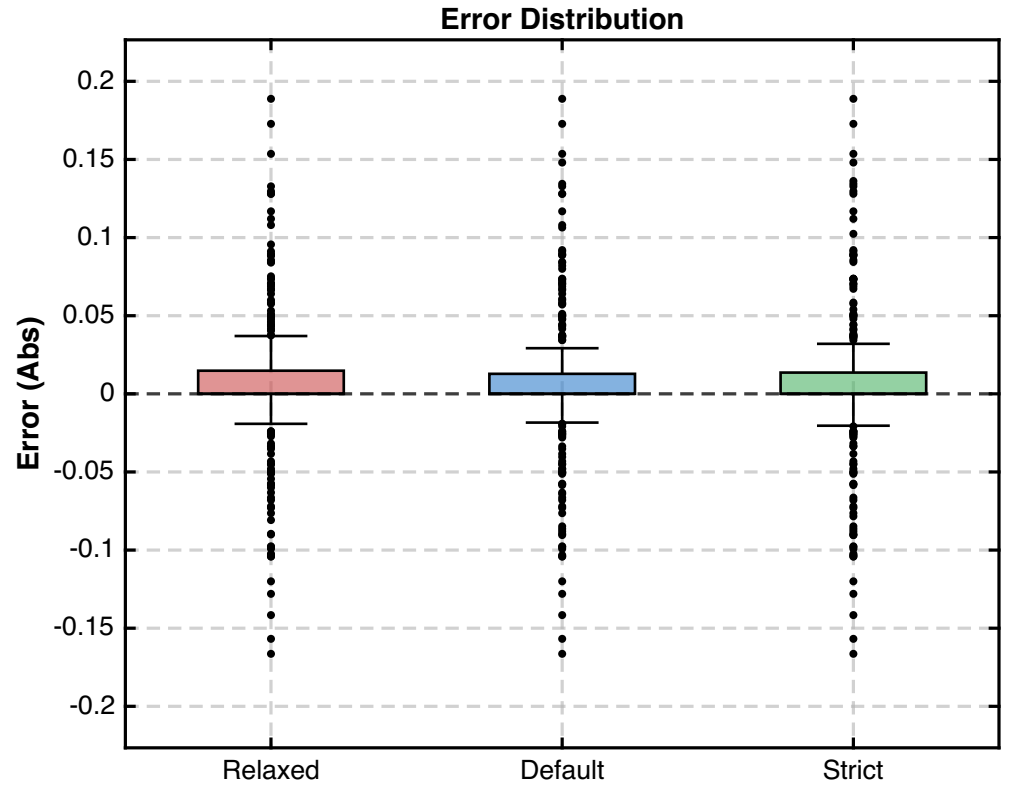
	Scenario Name	n	Distribution	Data Summary
1	**n = 25 (Normal)**	25	Normal	6 Datasets, Means 10-15, SD = 2.0
2	**n = 50 (Normal)**	50	Normal	6 Datasets, Means 10-15, SD = 2.0
3	**n = 100 (Normal)**	100	Normal	6 Datasets, Means 10-15, SD = 2.0
4	**n = 50 (Skewed)**	50	LogNormal	6 Datasets, Means 2.0-2.5 (Log), SD = 0.4 (Log)
5	**n = 50 (Likert)**	50	Likert	6 Datasets, Scale 1-7, Means 3-5, SD = 1.5
6	**n = 50 (Bimodal)**	50	Bimodal	6 Datasets, Mix Means 10 & 15, SD = 2.7
7	**n = 50 (Large Effect)**	50	NormalLarge	6 Datasets, Means 10-20.0, SD = 2.0

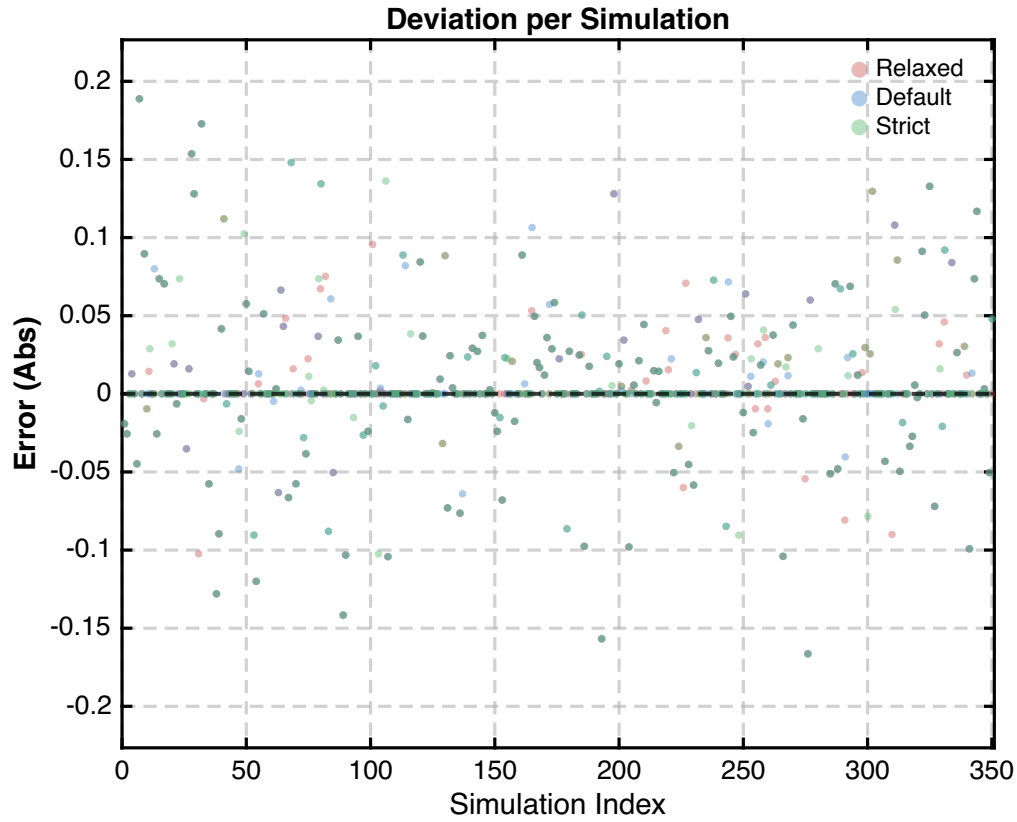
Configuration: Method Parameters

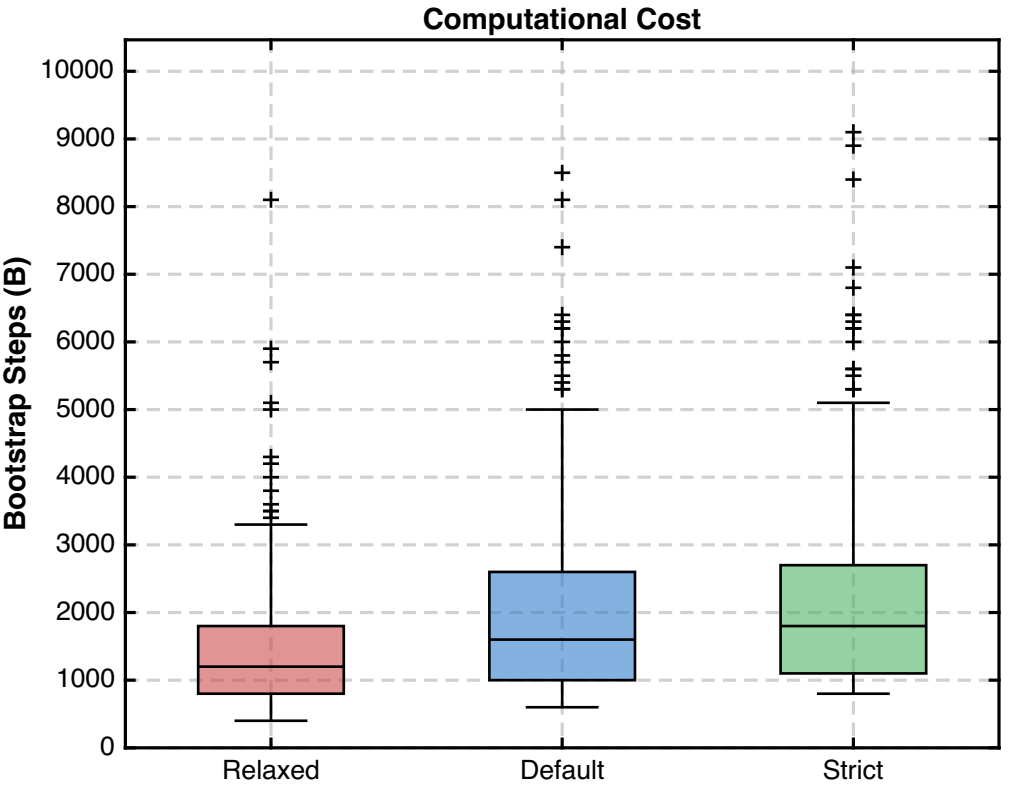
References (B): Thresholds = 25000 | BCa = 50000 | Ranking = 10000

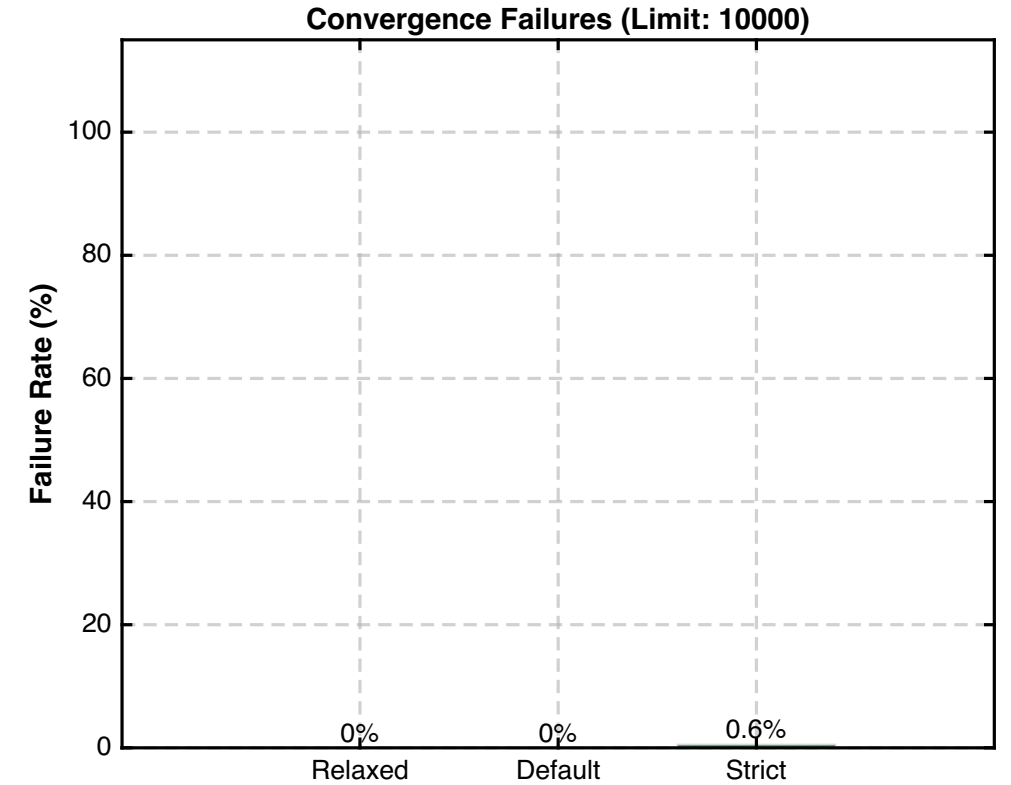
Parameter Set	Relaxed	Default	Strict
1. Bootstrap Thresholds			
Trials (n)	15	25	35
Smoothing (sm)	2	3	4
Streak (st)	2	3	4
Tolerance (tol)	0.010	0.010	0.010
Step Size (B)	100	100	100
B Range	[100-10000]	[100-10000]	[100-10000]
2. BCa Confidence Intervals			
Trials (n)	20	30	40
Smoothing (sm)	2	3	4
Streak (st)	2	3	4
Tolerance (tol)	0.030	0.030	0.030
Step Size (B)	200	200	200
B Range	[100-20000]	[100-20000]	[100-20000]
3. Ranking Stability			
Trials (n)	10	15	20
Smoothing (sm)	2	3	4
Streak (st)	2	3	4
Tolerance (tol)	0.005	0.005	0.005
Step Size (B)	25	25	25
B Range	[50-2500]	[50-2500]	[50-2500]

Thresholds (Delta) Analysis: Global Summary (Ref B = 25000)

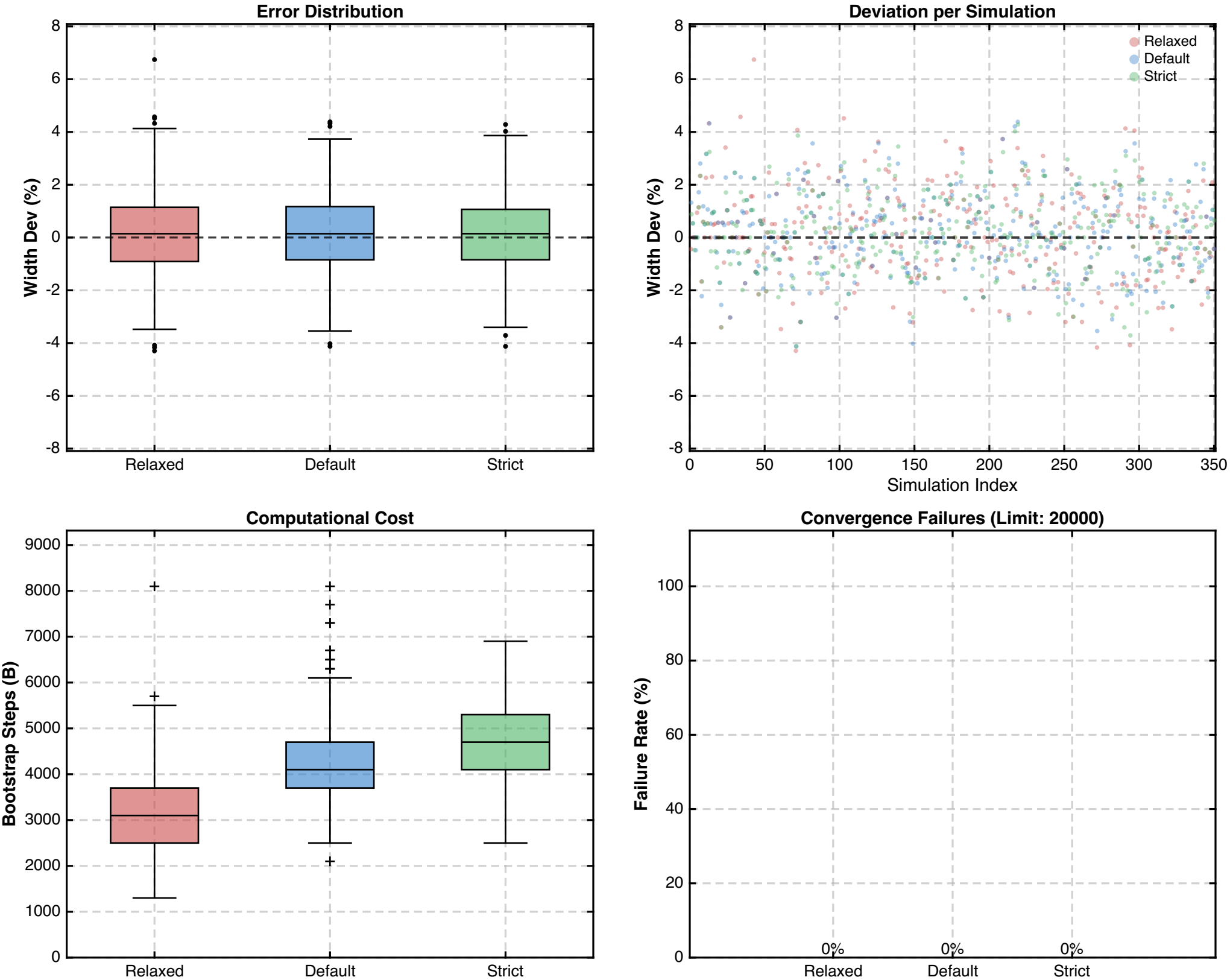




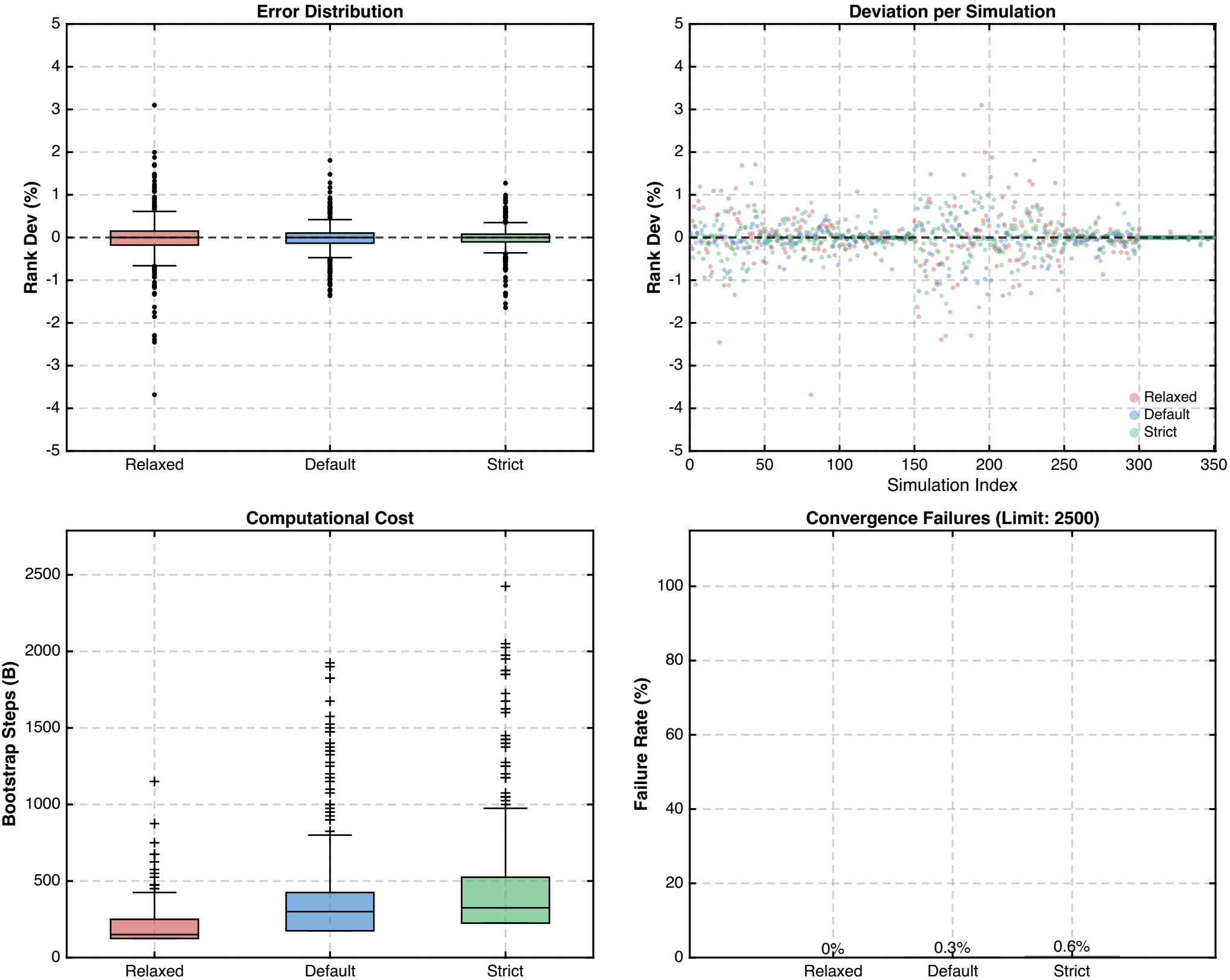




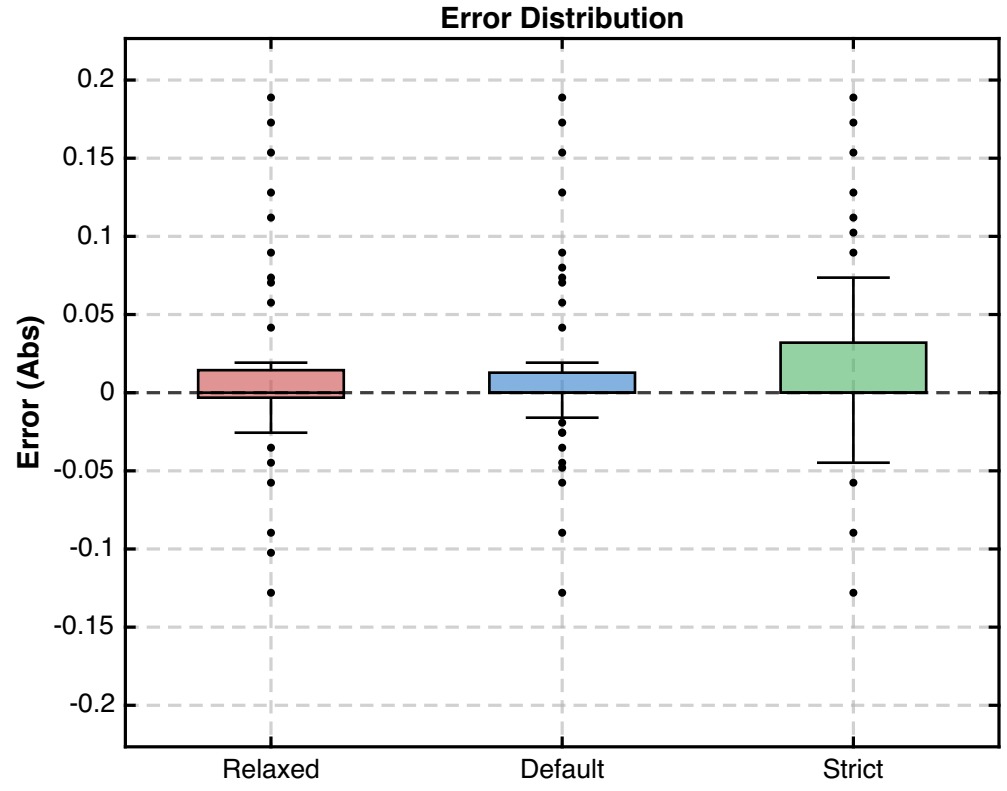
BCa CI (Width) Analysis: Global Summary (Ref B = 50000)

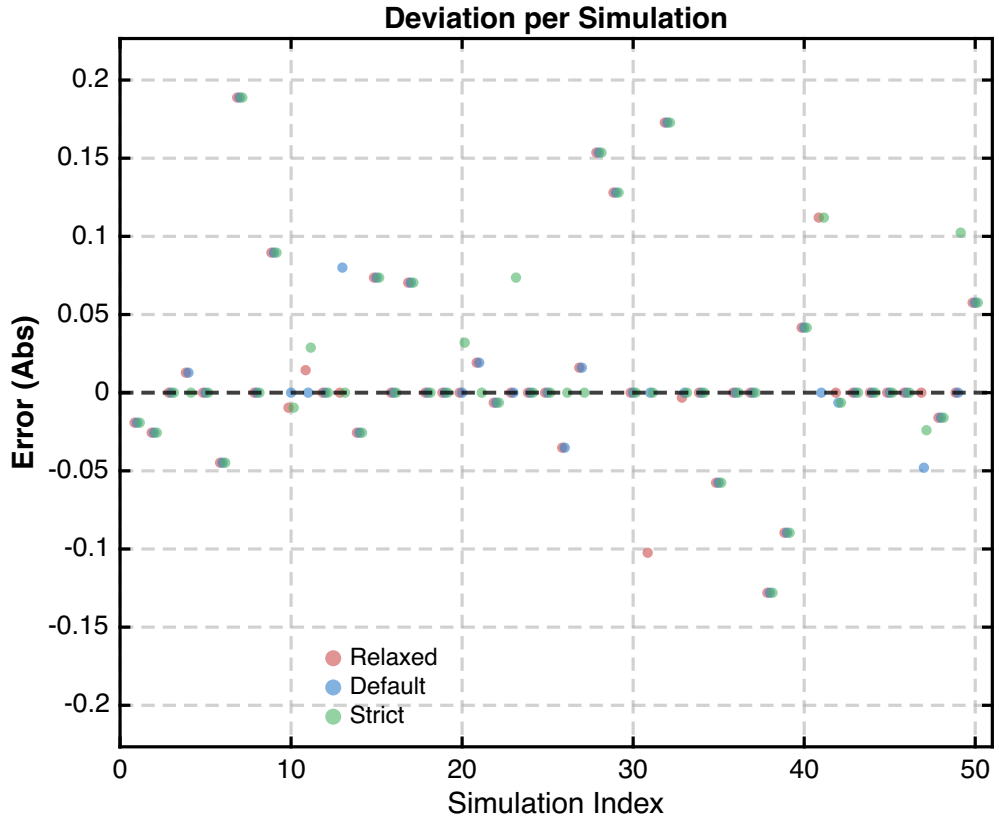


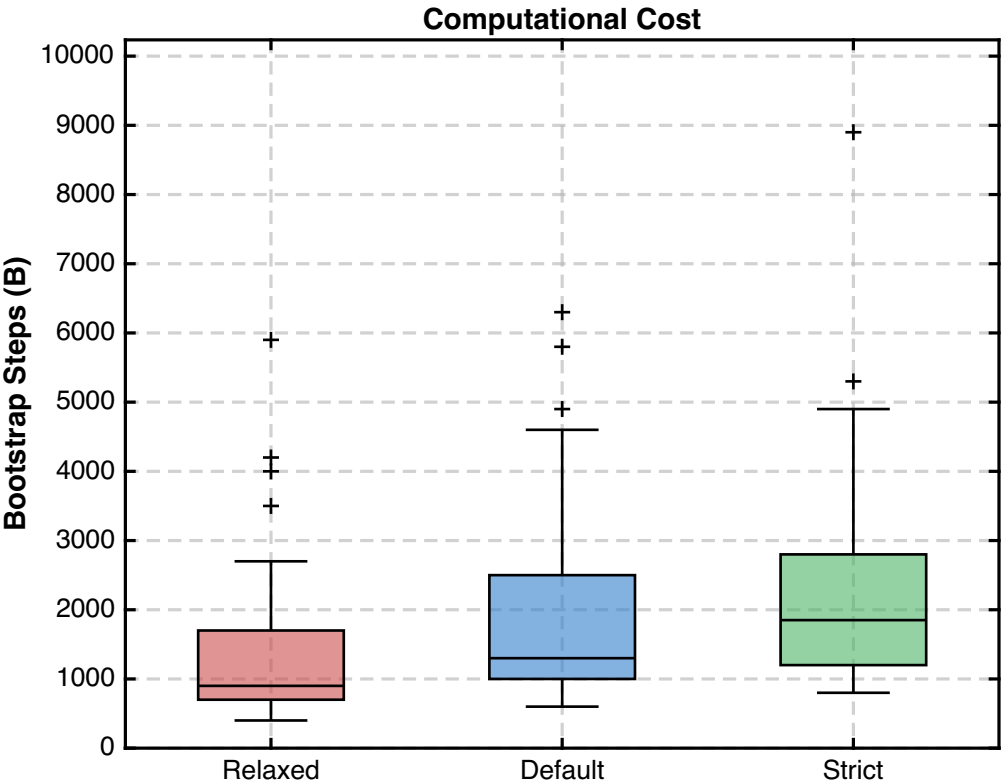
Ranking (Mean) Analysis: Global Summary (Ref B = 10000)

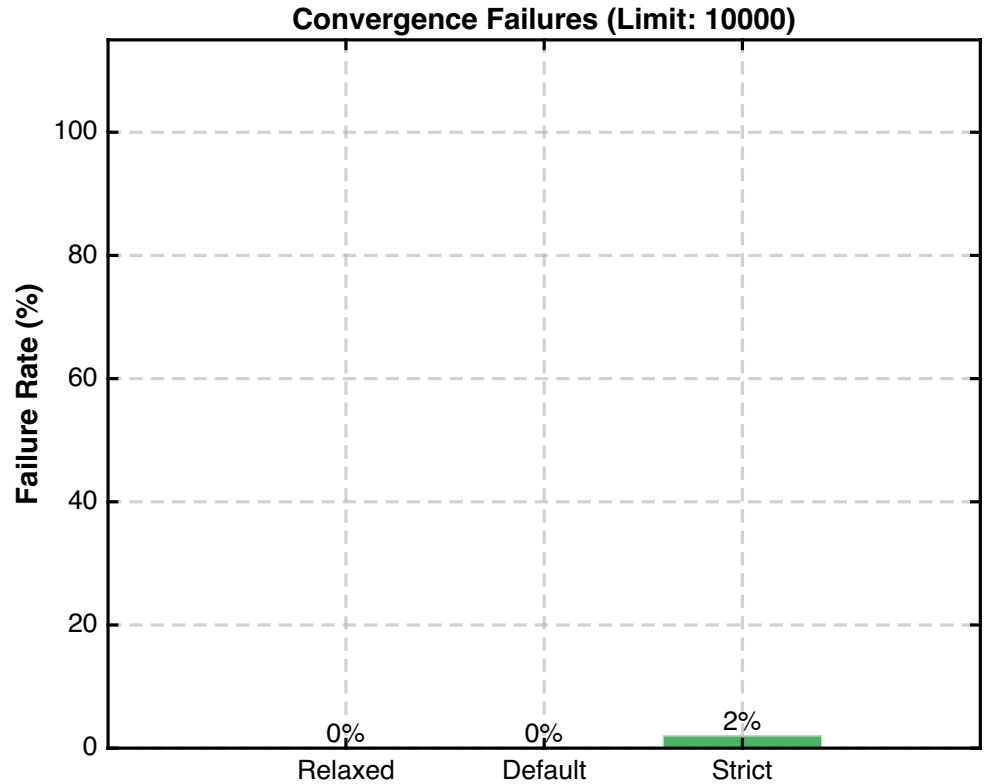


Thresholds (Delta) Analysis: n 25 Normal (Median Cliff's d = 0.56)

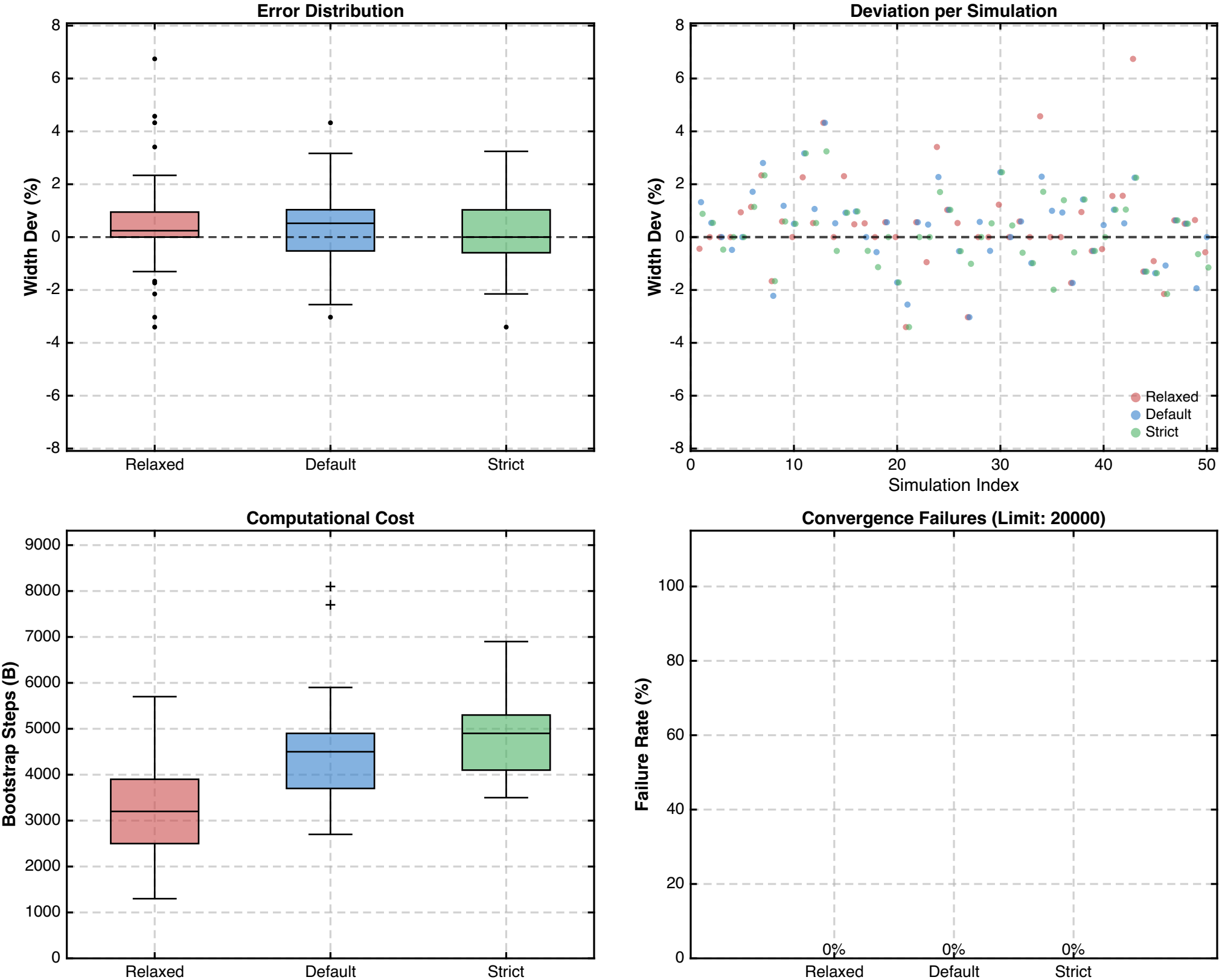




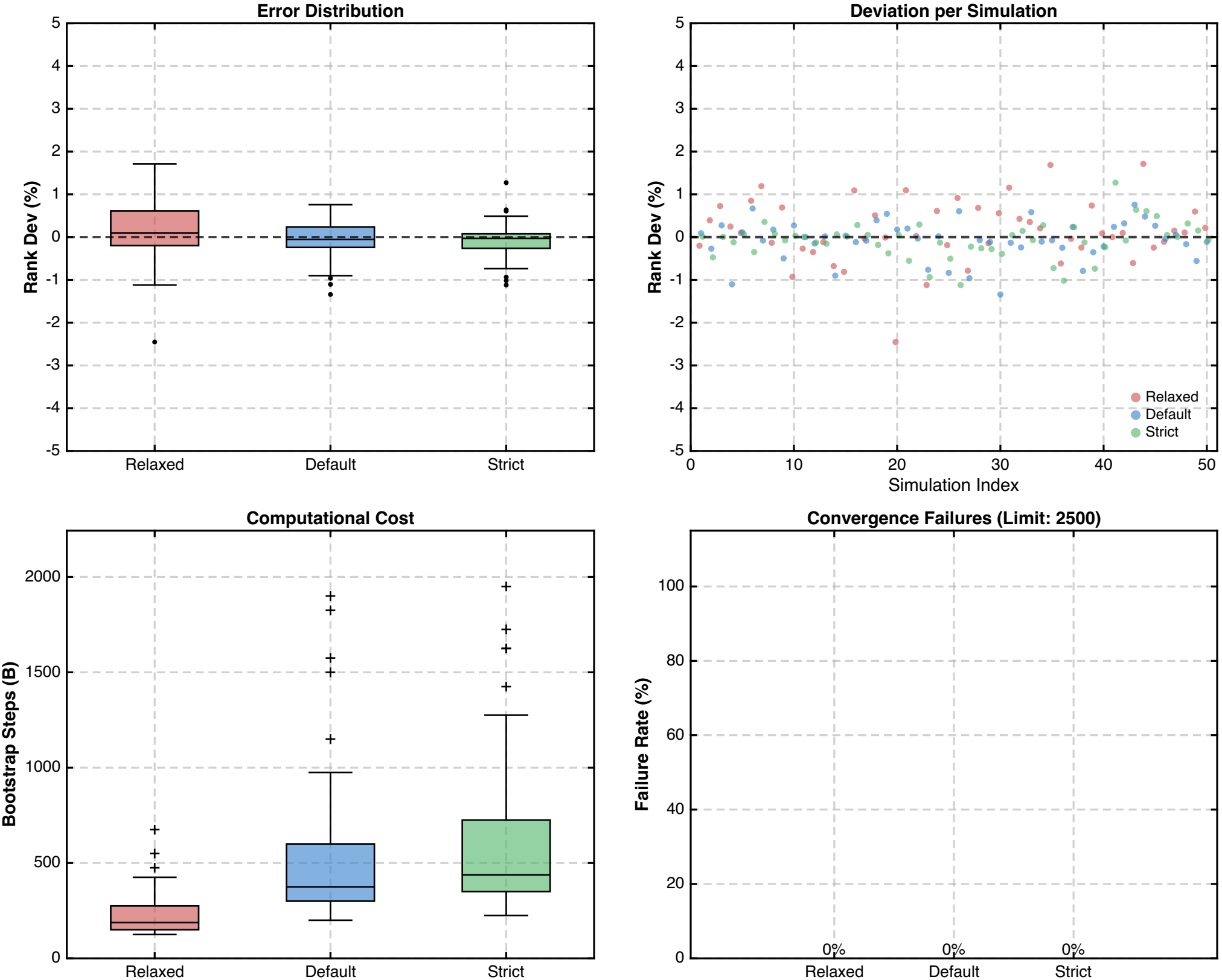




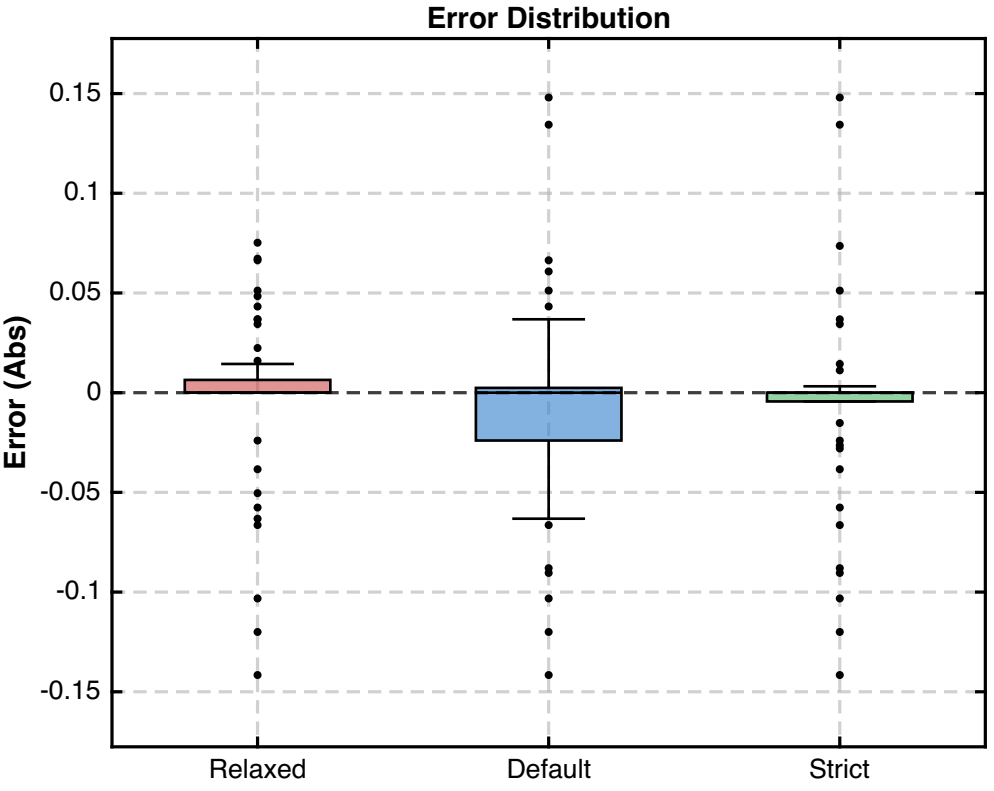
BCa CI (Width) Analysis: n 25 Normal (Median Cliff's d = 0.56)

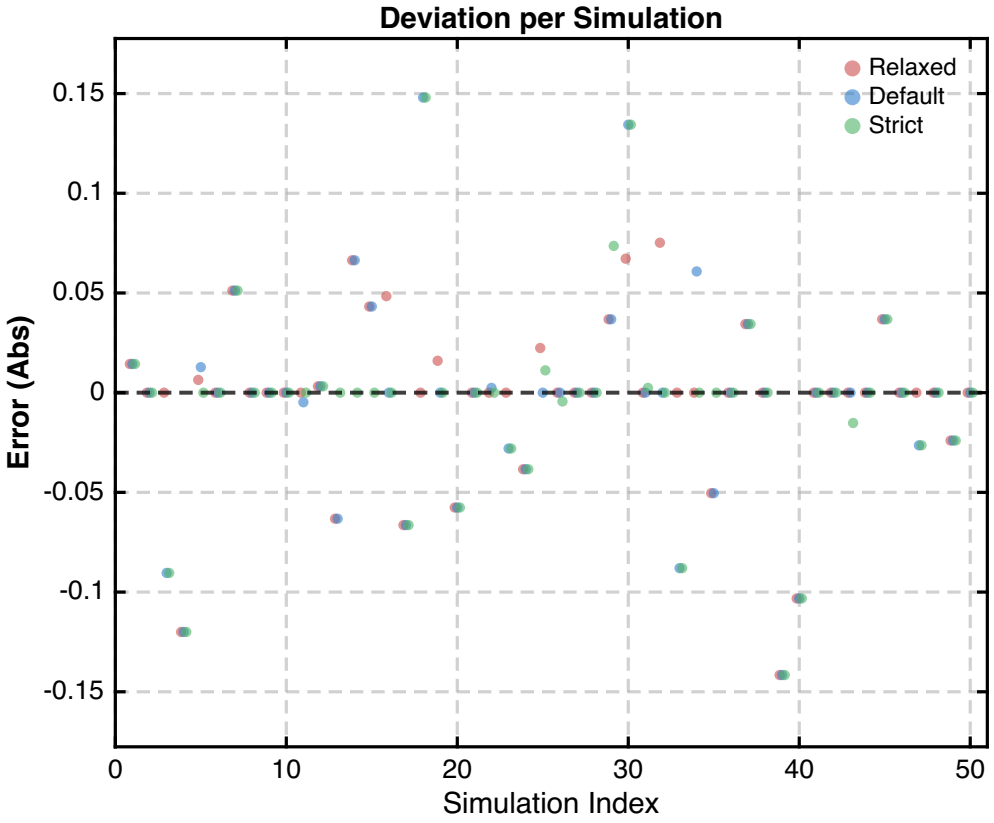


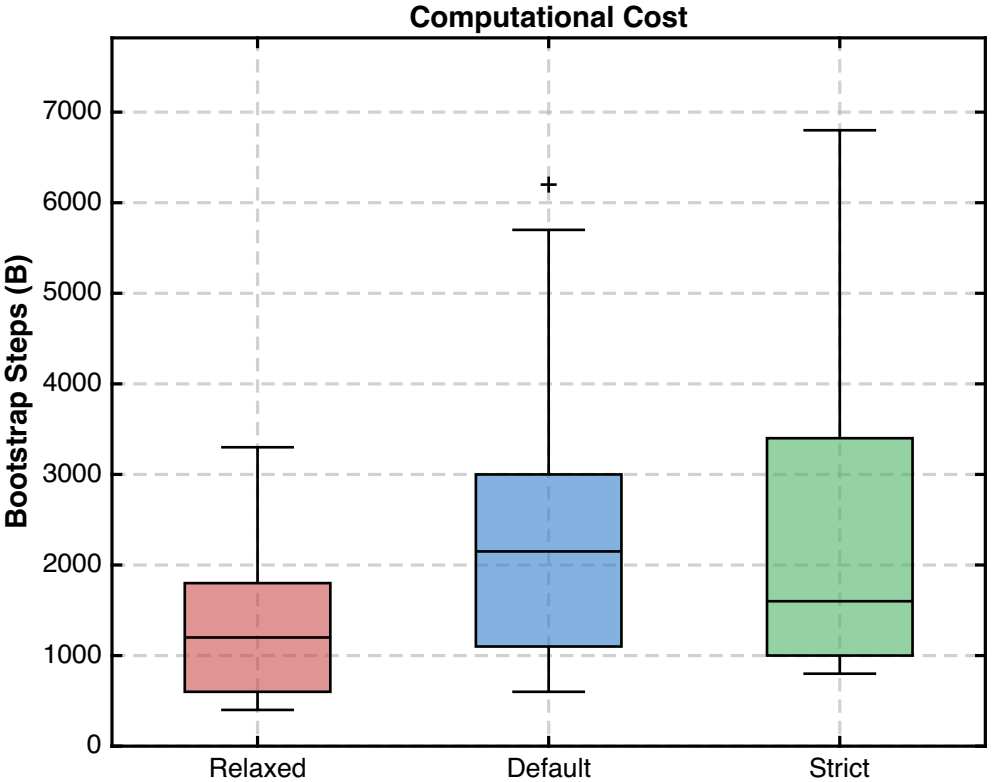
Ranking (Mean) Analysis: n 25 Normal (Median Cliff's d = 0.56)

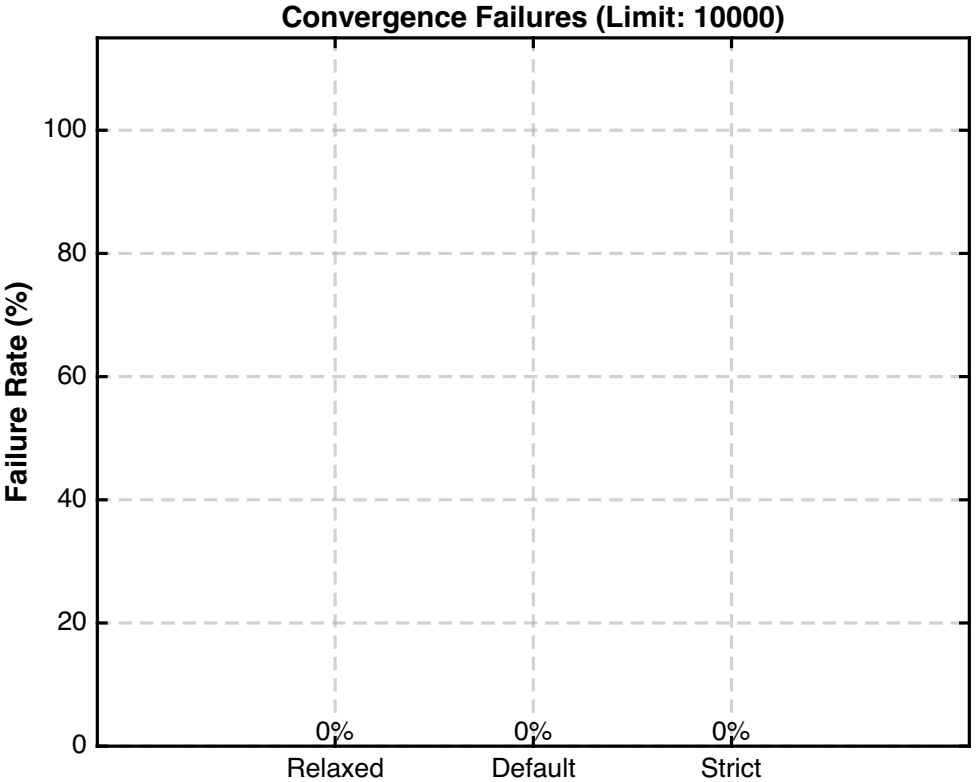


Thresholds (Delta) Analysis: n 50 Normal (Median Cliff's d = 0.54)

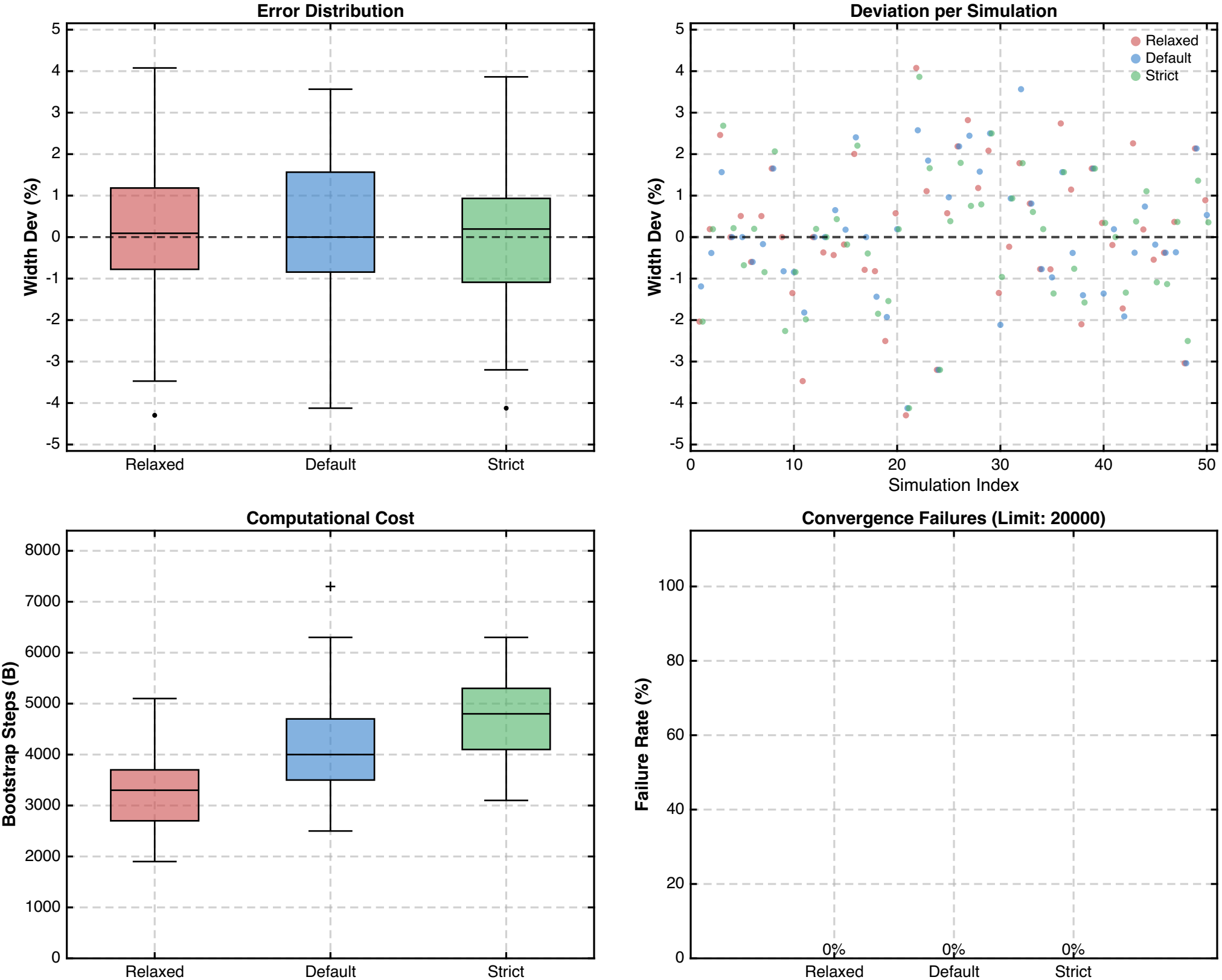




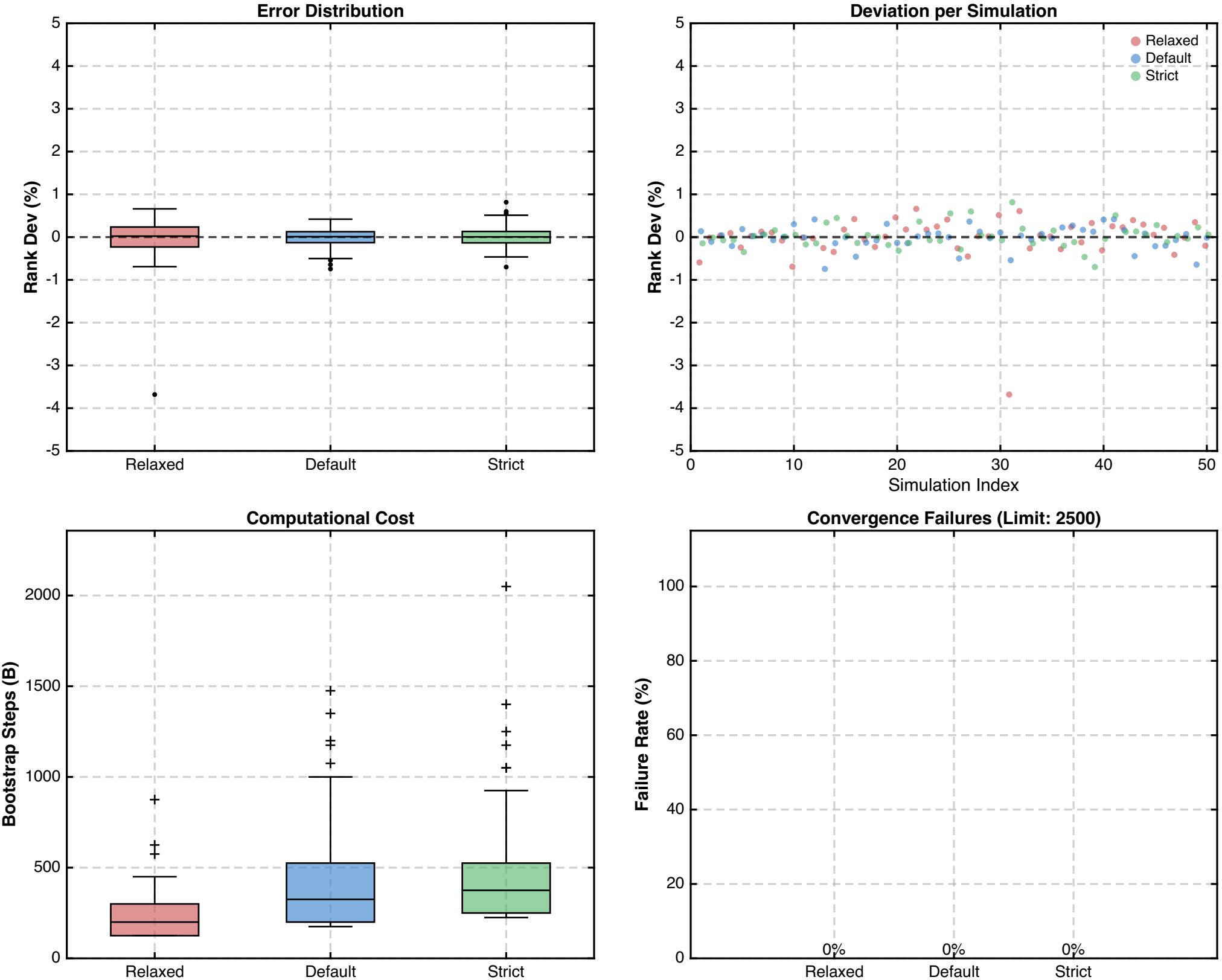




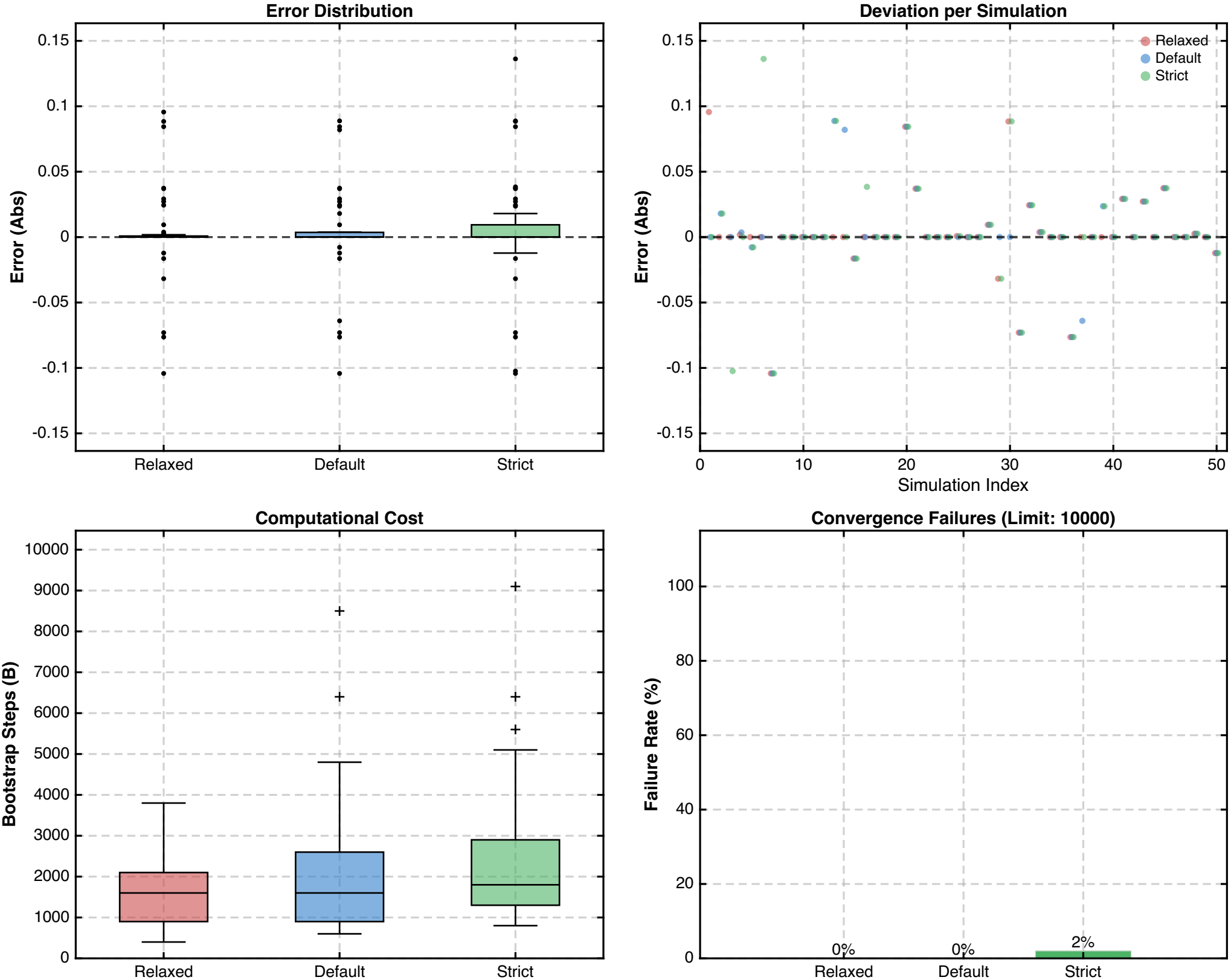
BCa CI (Width) Analysis: n 50 Normal (Median Cliff's d = 0.54)



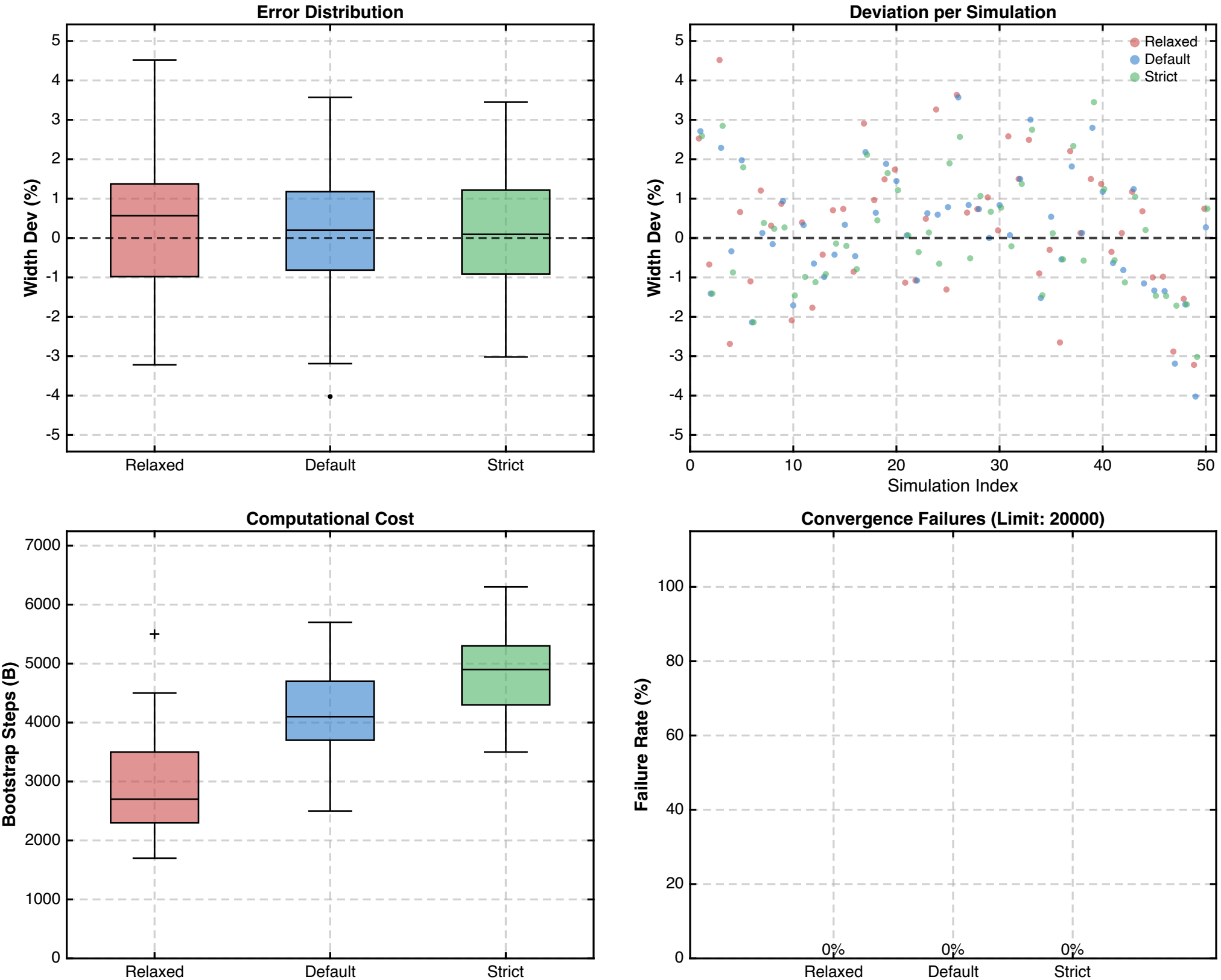
Ranking (Mean) Analysis: n 50 Normal (Median Cliff's d = 0.54)



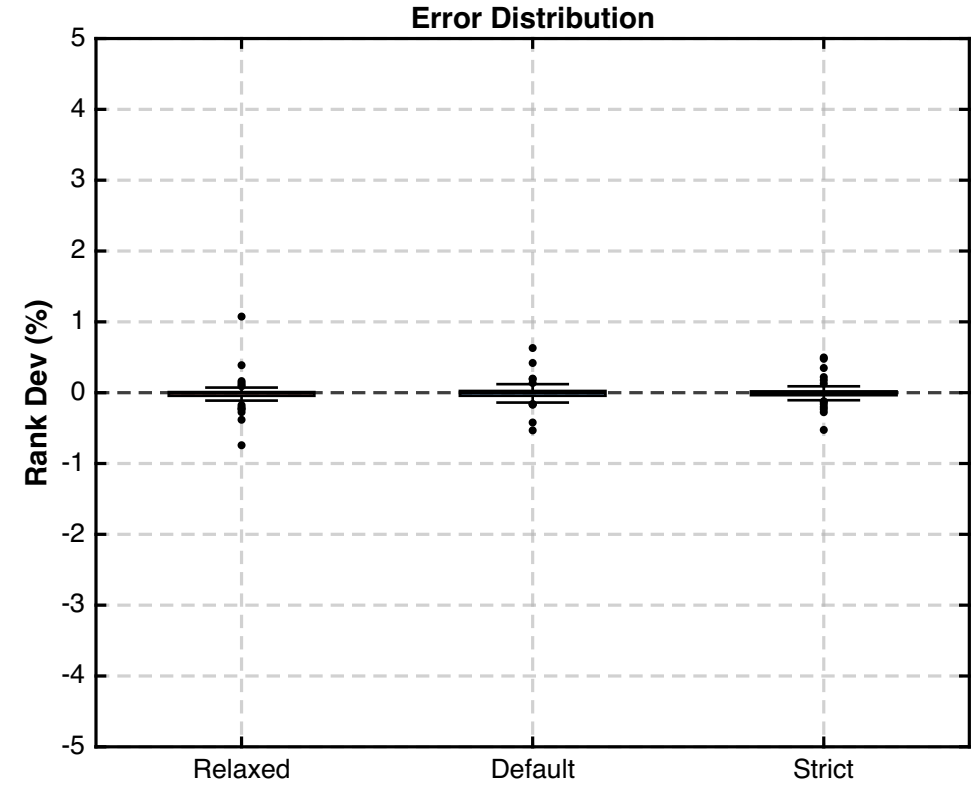
Thresholds (Delta) Analysis: n 100 Normal (Median Cliff's d = 0.54)

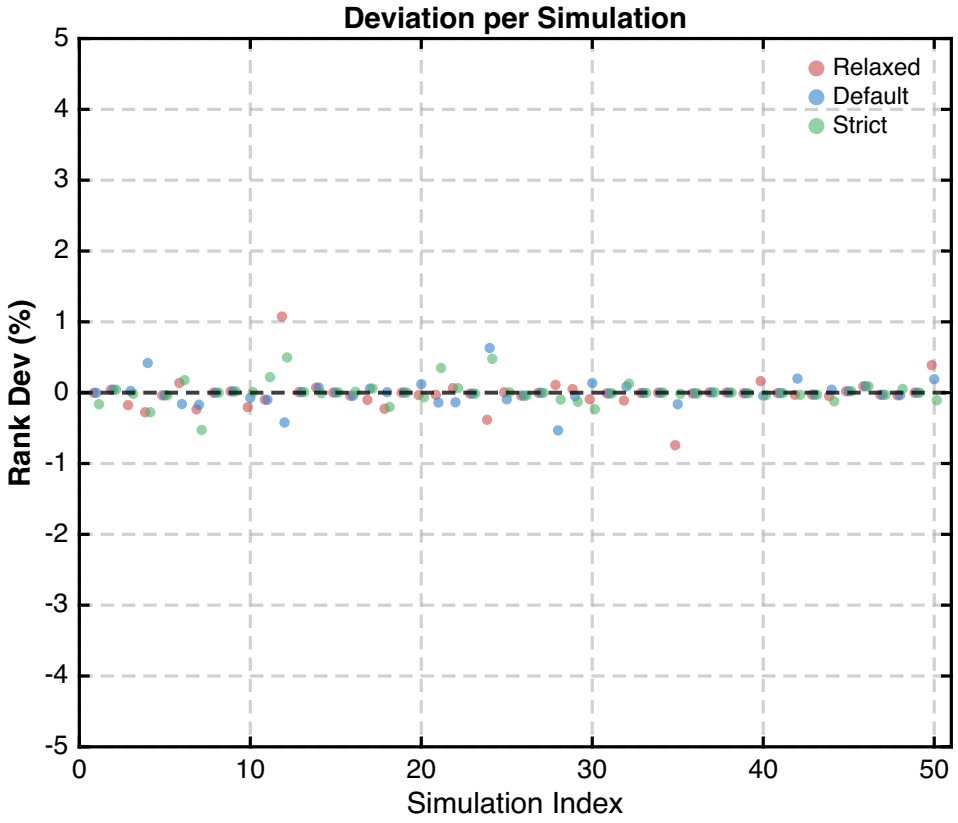


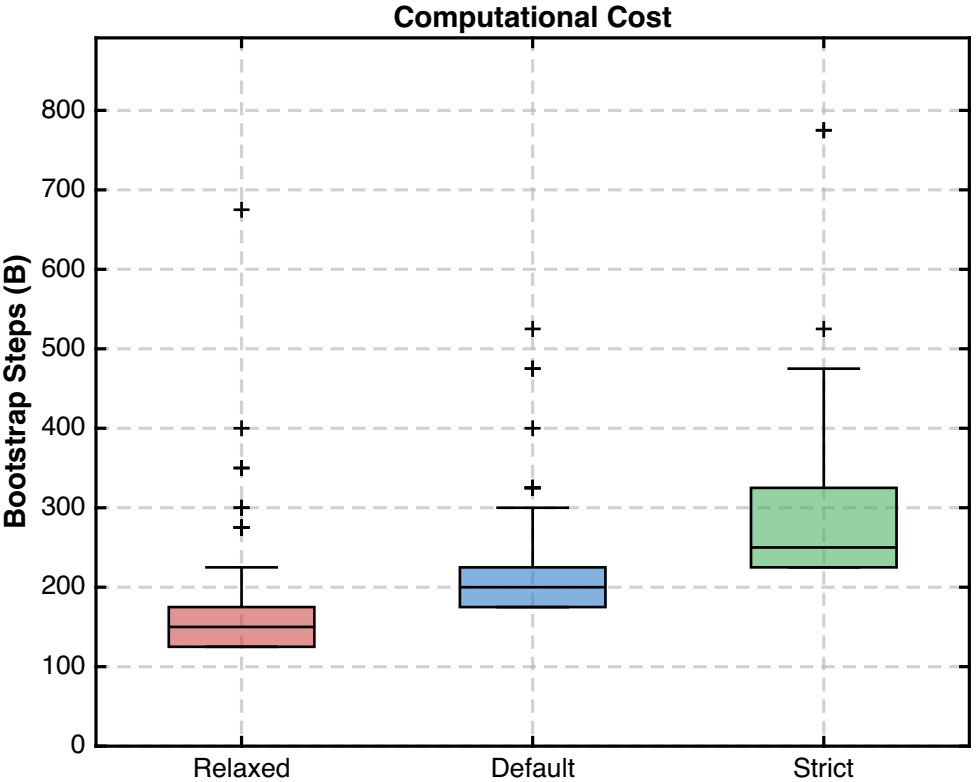
BCa CI (Width) Analysis: n 100 Normal (Median Cliff's d = 0.54)

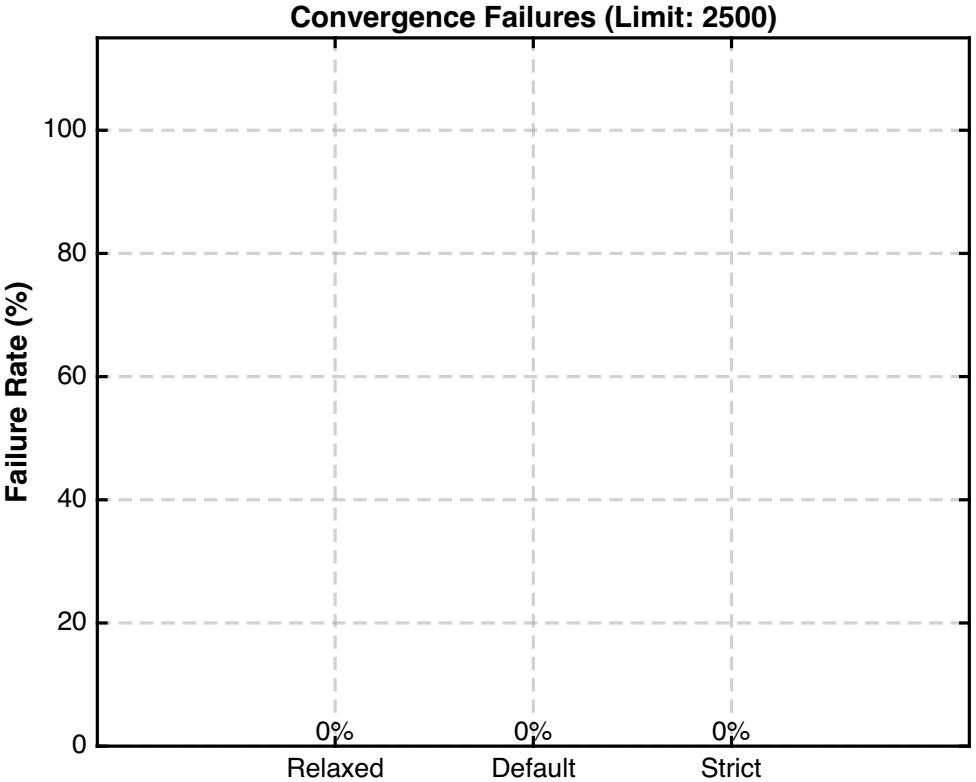


Ranking (Mean) Analysis: n 100 Normal (Median Cliff's d = 0.54)

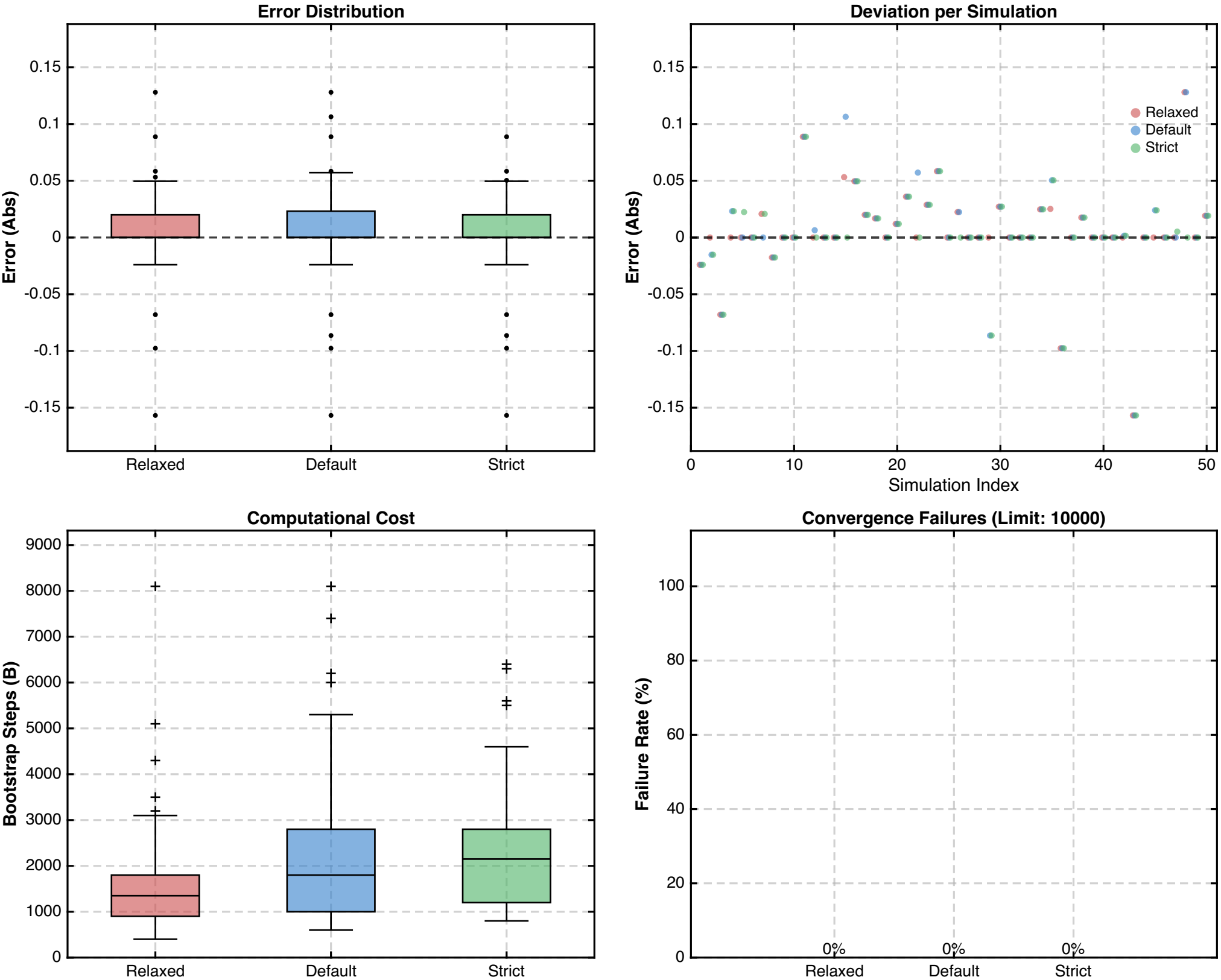




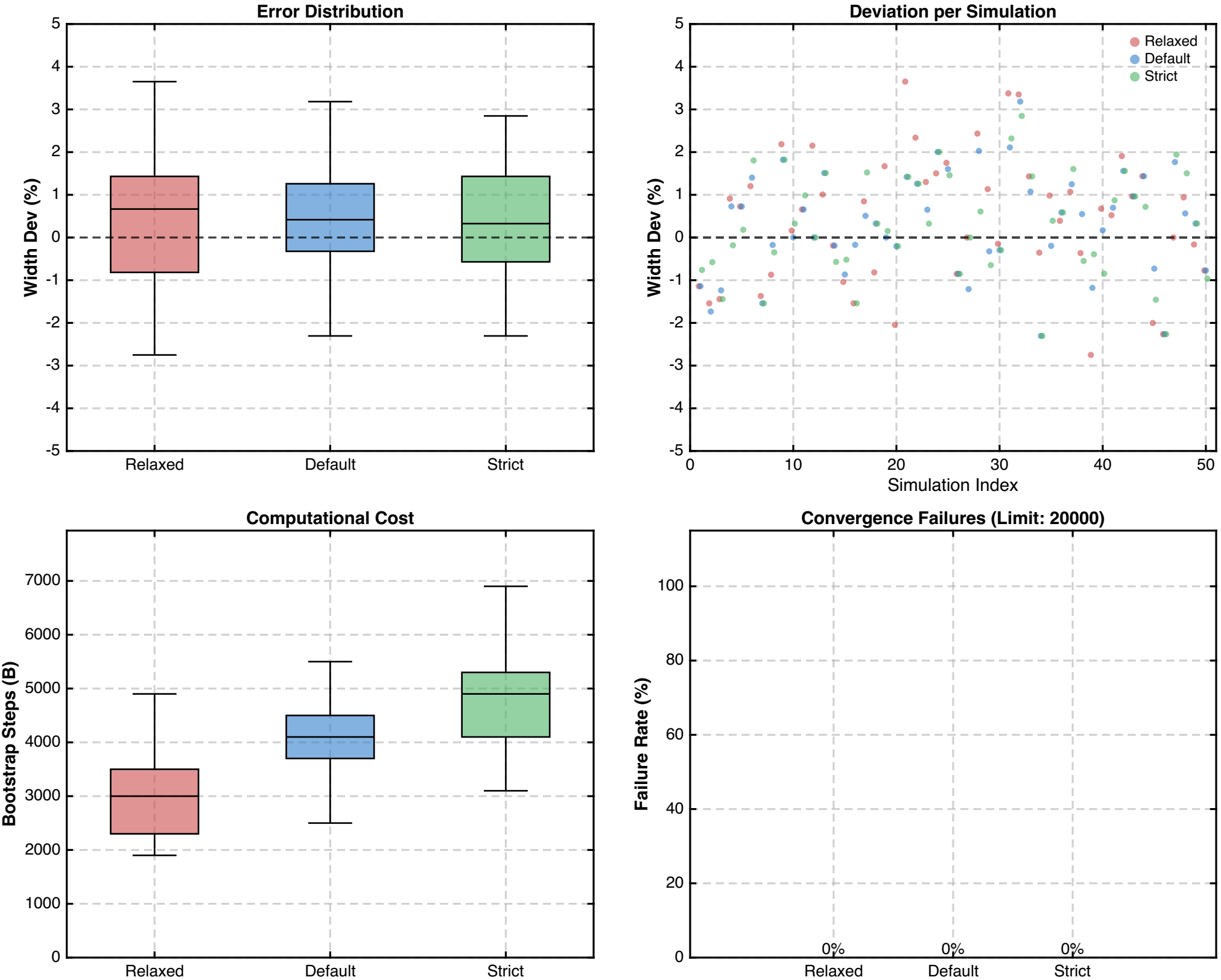




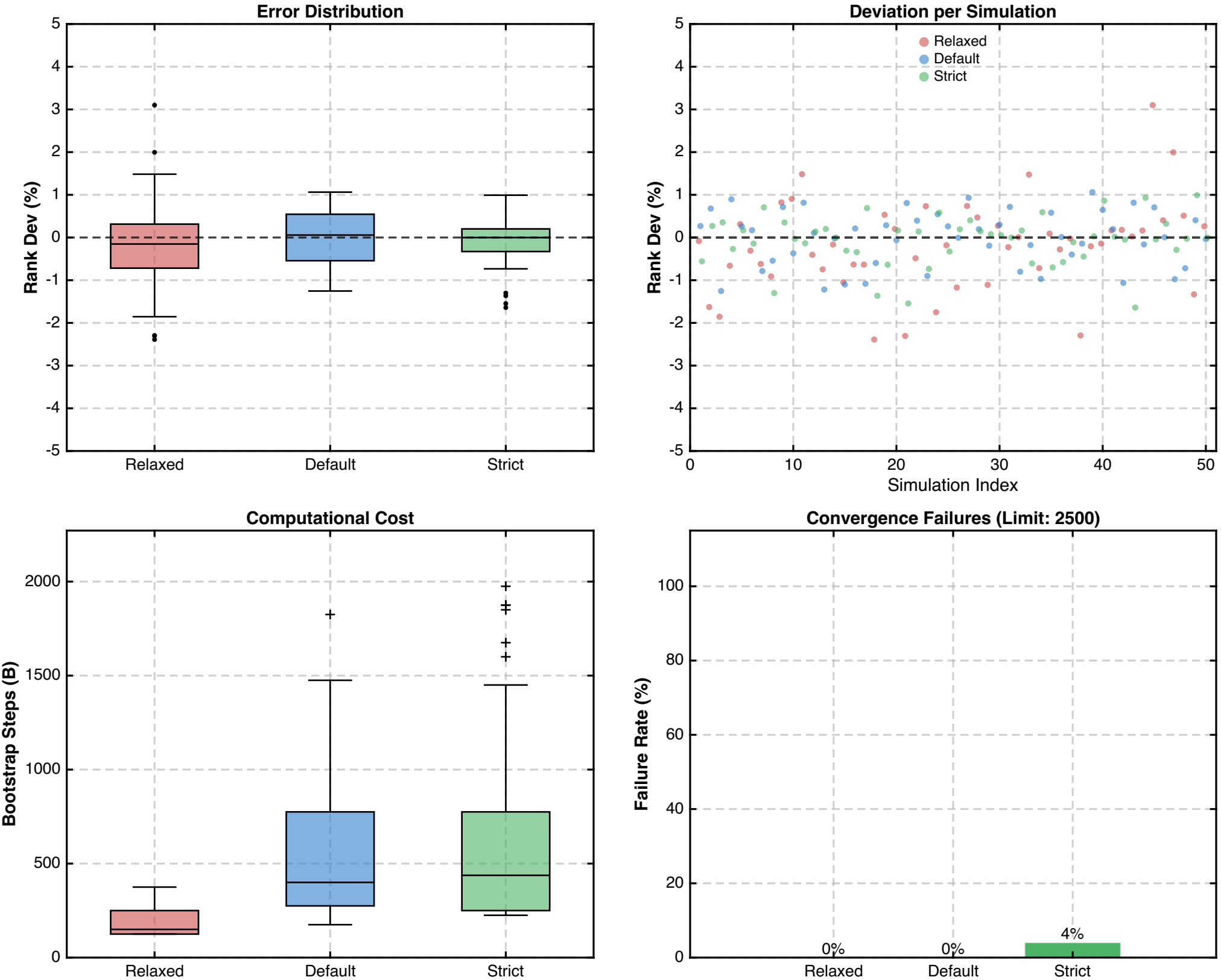
Thresholds (Delta) Analysis: n 50 Skewed (Median Cliff's d = 0.29)



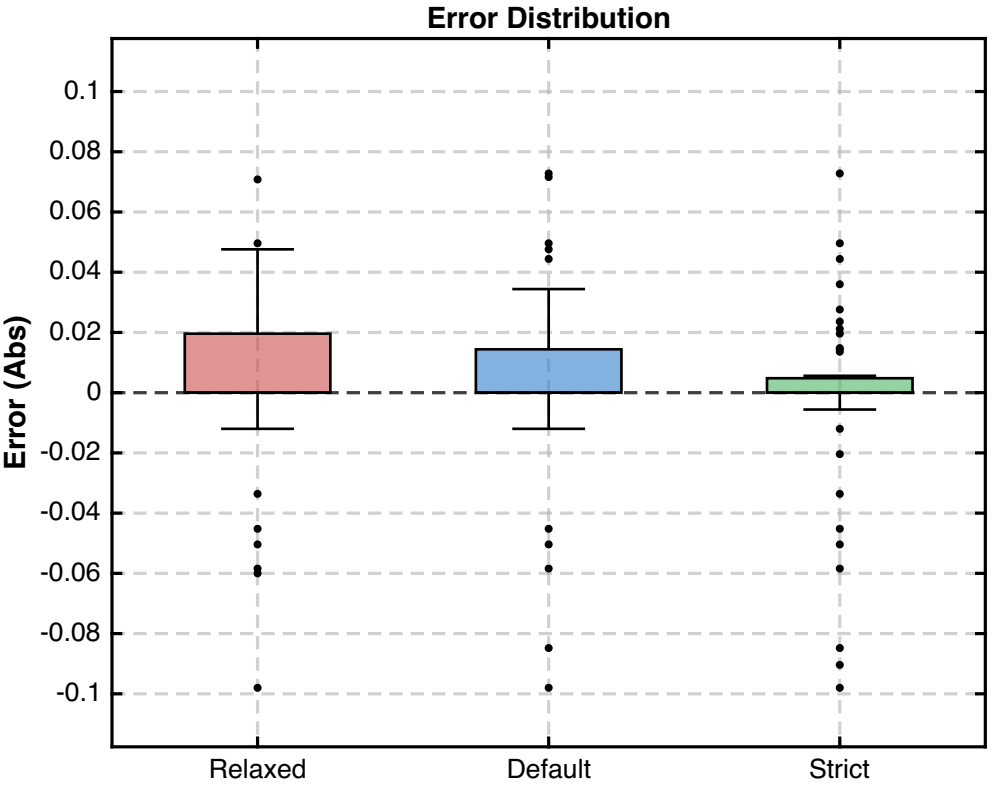
BCa CI (Width) Analysis: n 50 Skewed (Median Cliff's d = 0.29)

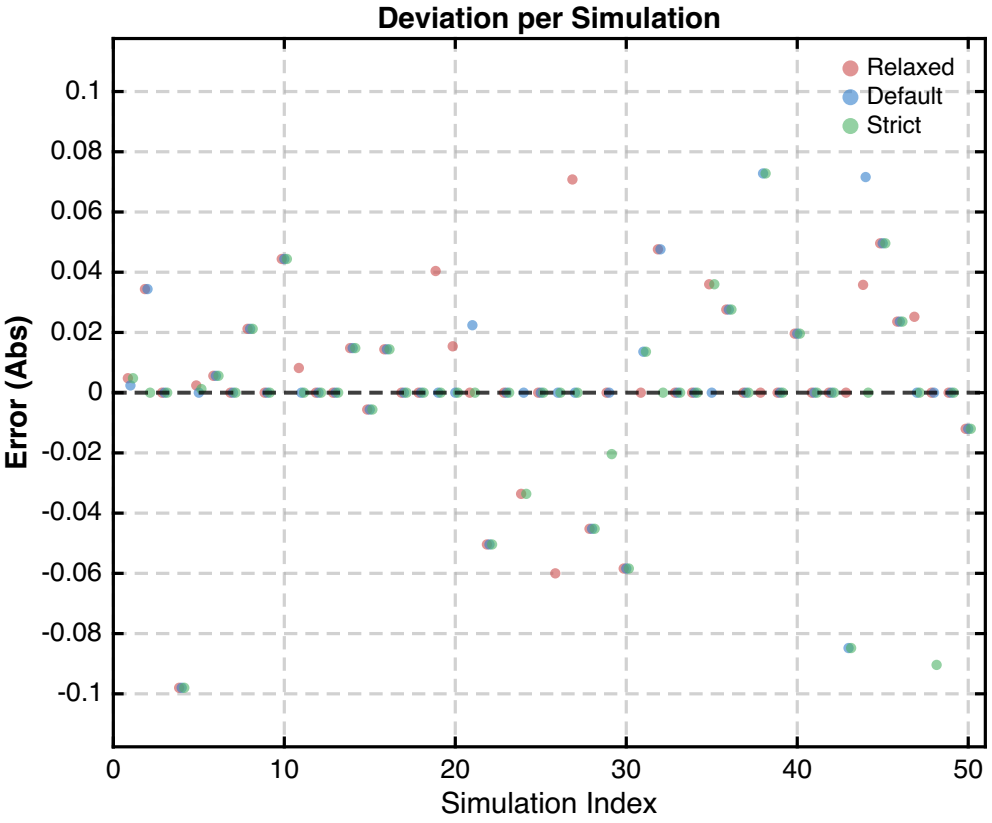


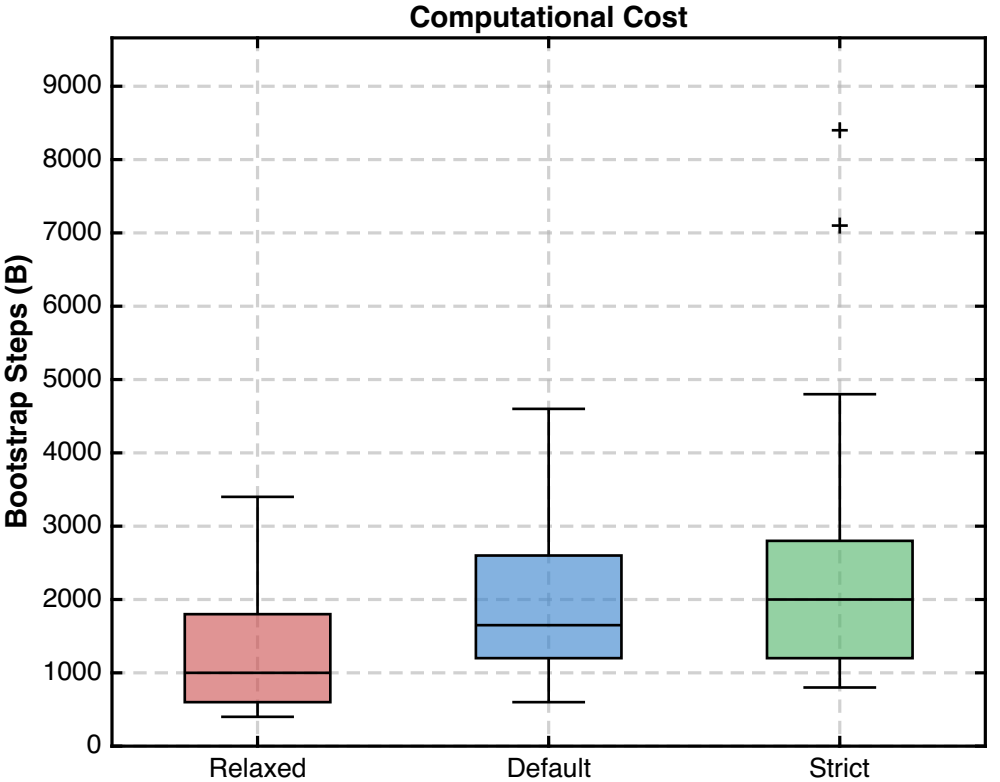
Ranking (Mean) Analysis: n 50 Skewed (Median Cliff's d = 0.29)

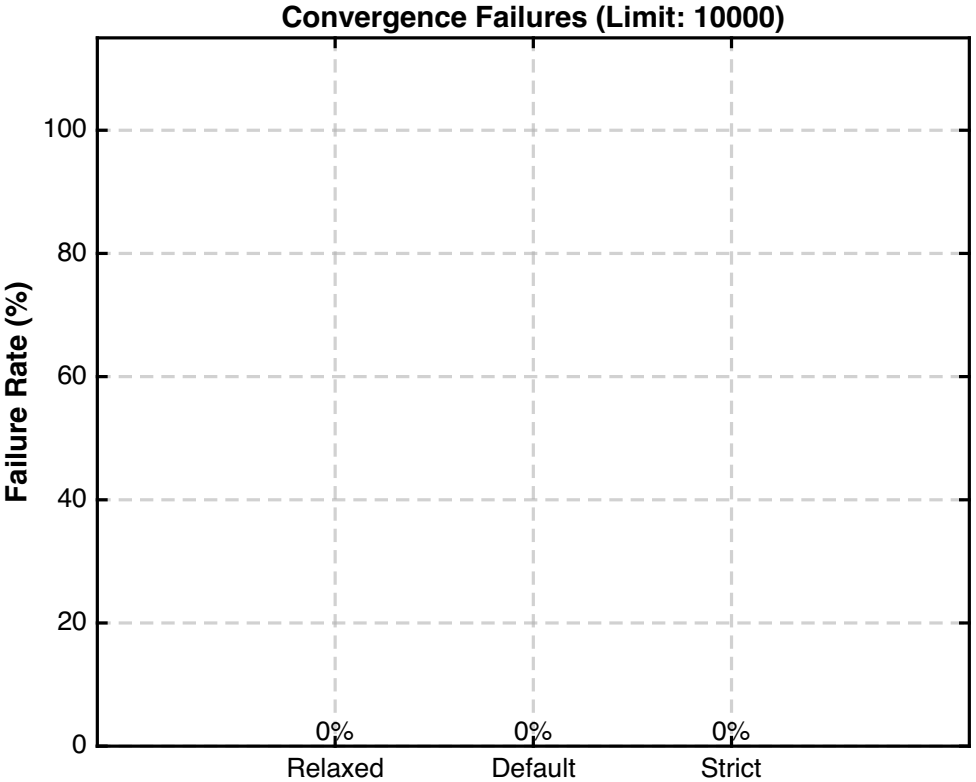


Thresholds (Delta) Analysis: n 50 Likert (Median Cliff's d = 0.31)

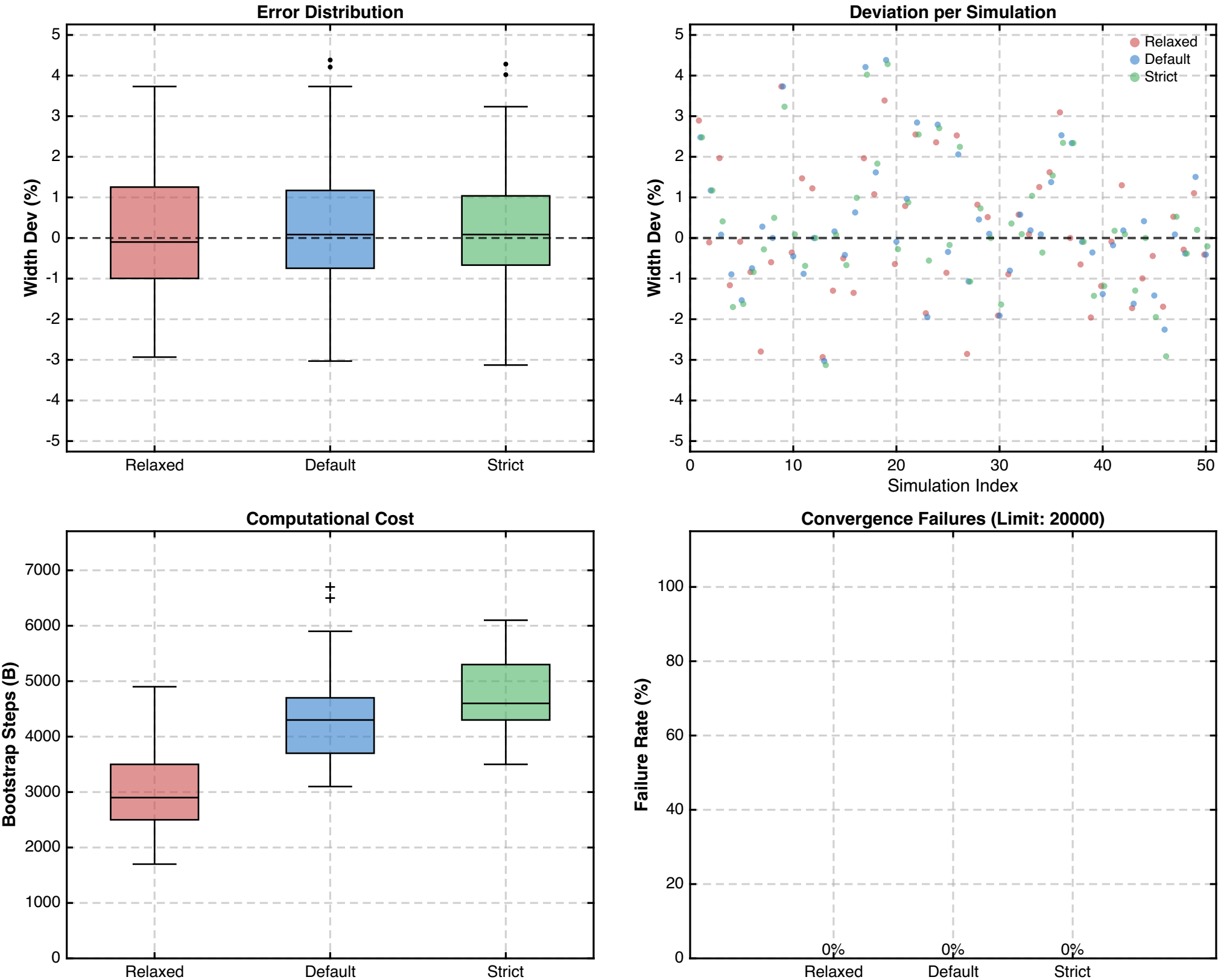




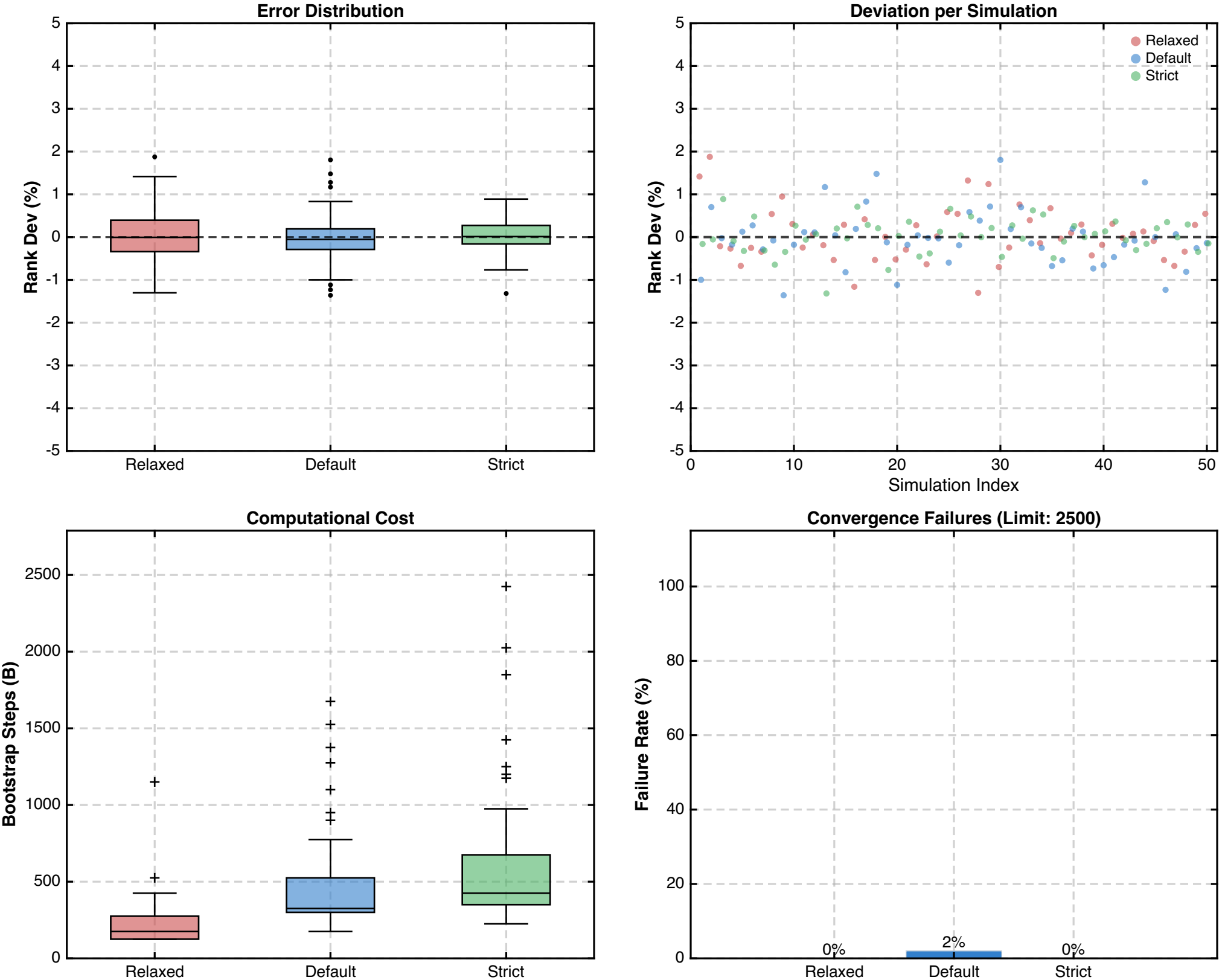




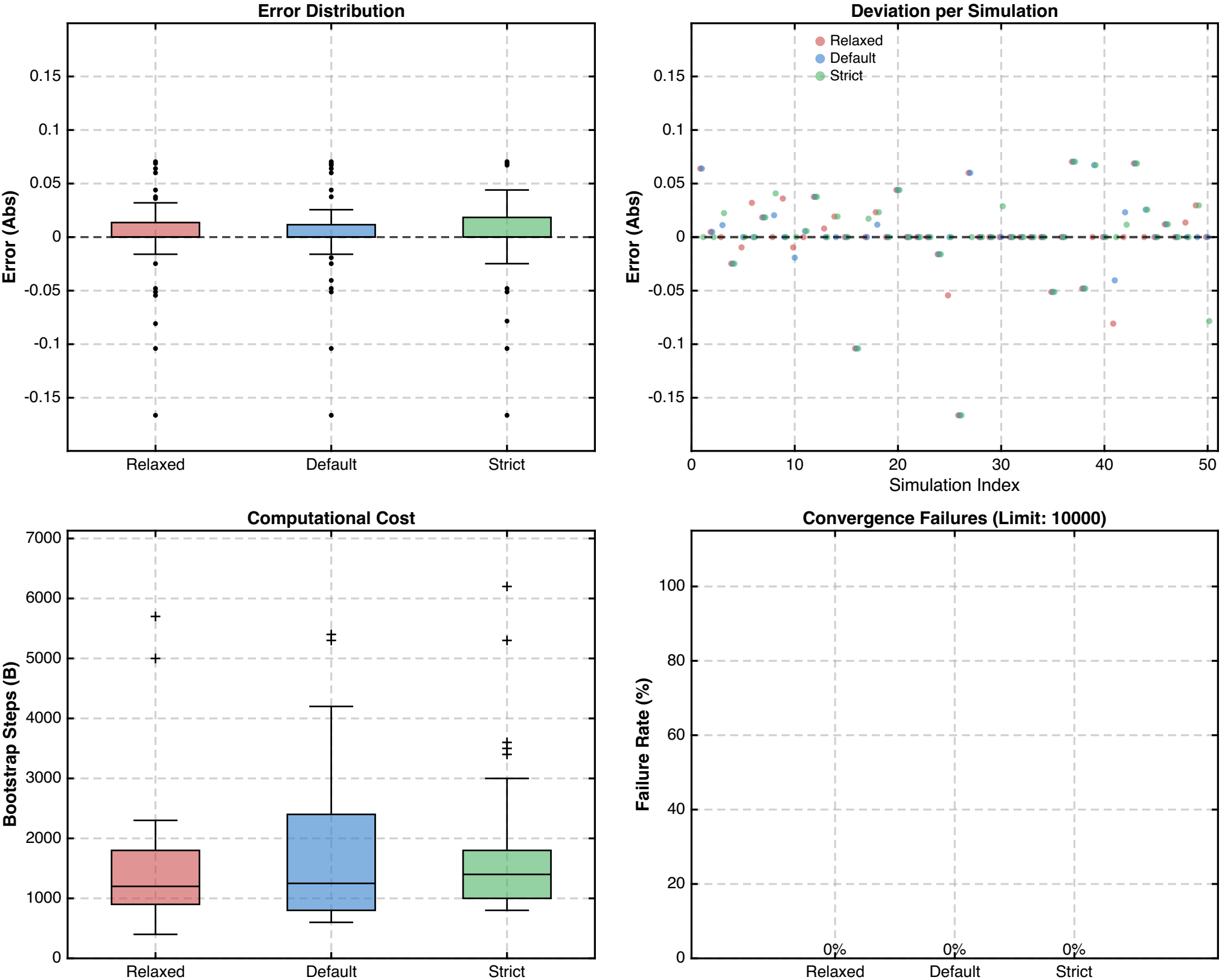
BCa CI (Width) Analysis: n 50 Likert (Median Cliff's d = 0.31)



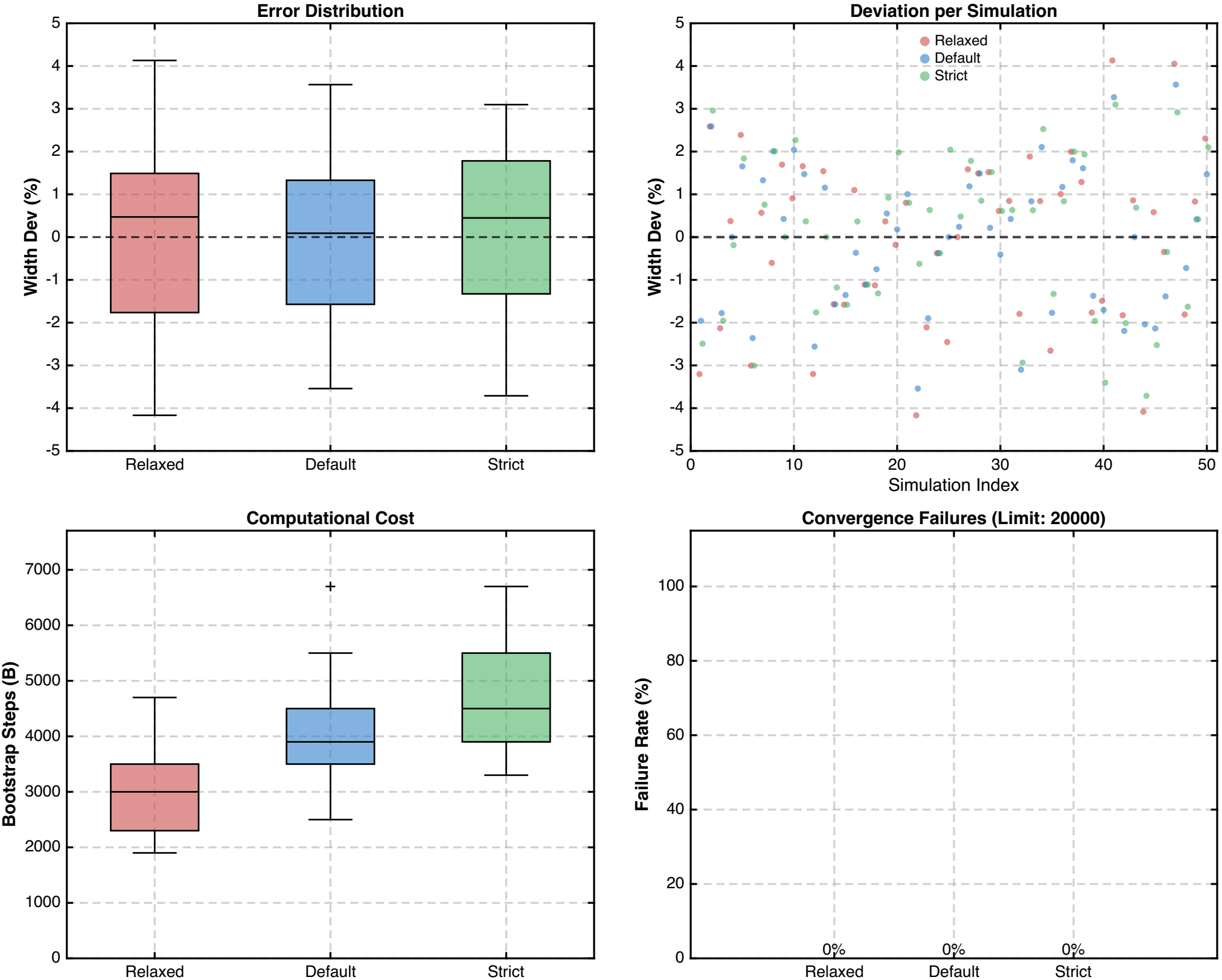
Ranking (Mean) Analysis: n 50 Likert (Median Cliff's d = 0.31)



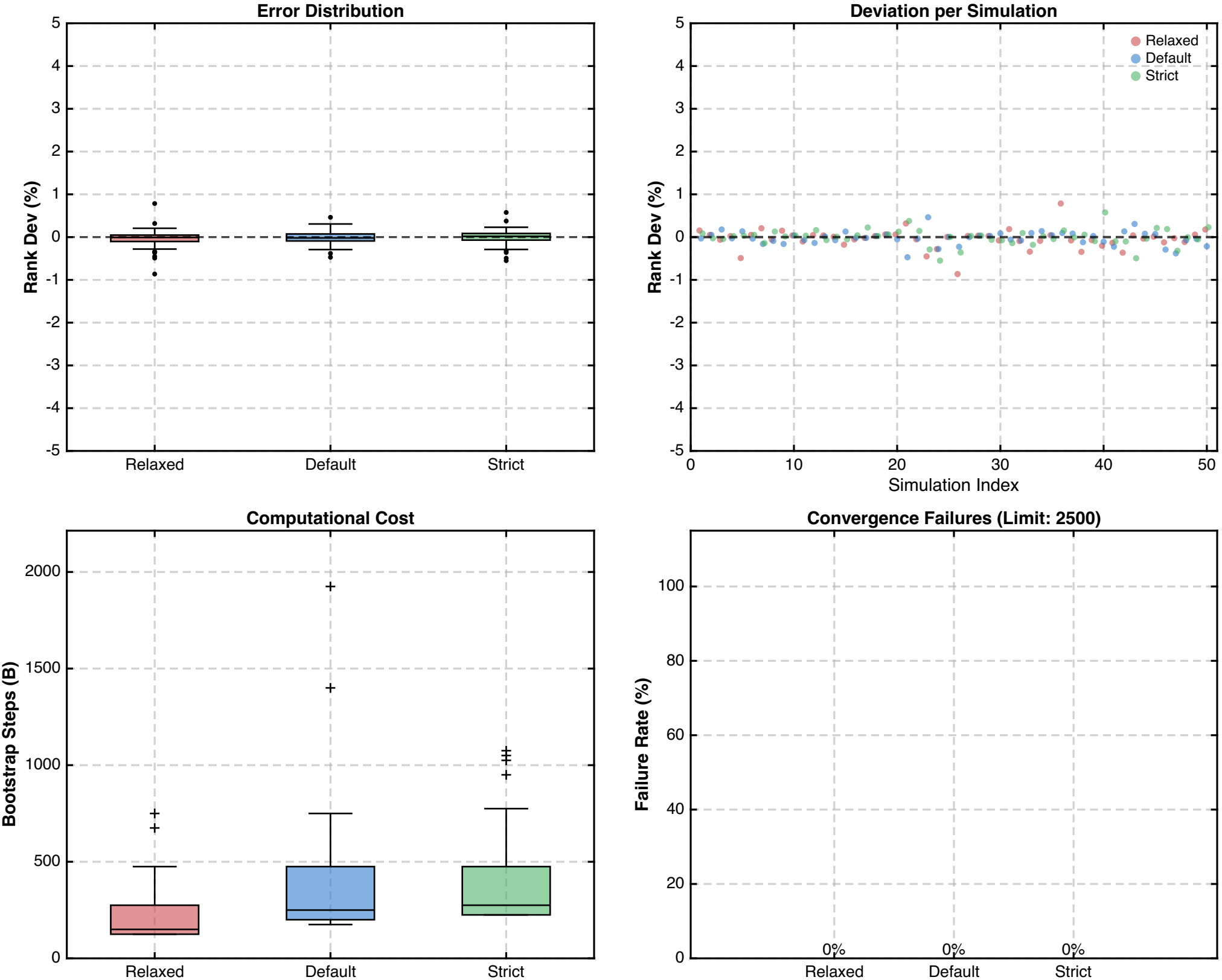
Thresholds (Delta) Analysis: n 50 Bimodal (Median Cliff's d = 0.56)



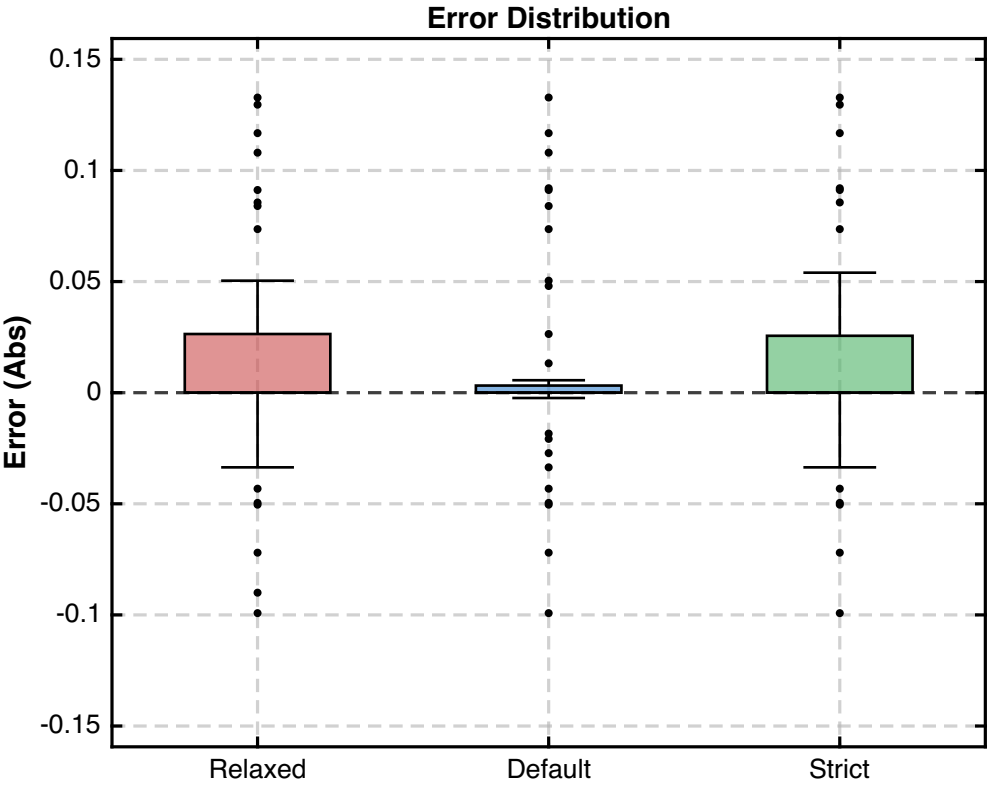
BCa CI (Width) Analysis: n 50 Bimodal (Median Cliff's d = 0.56)

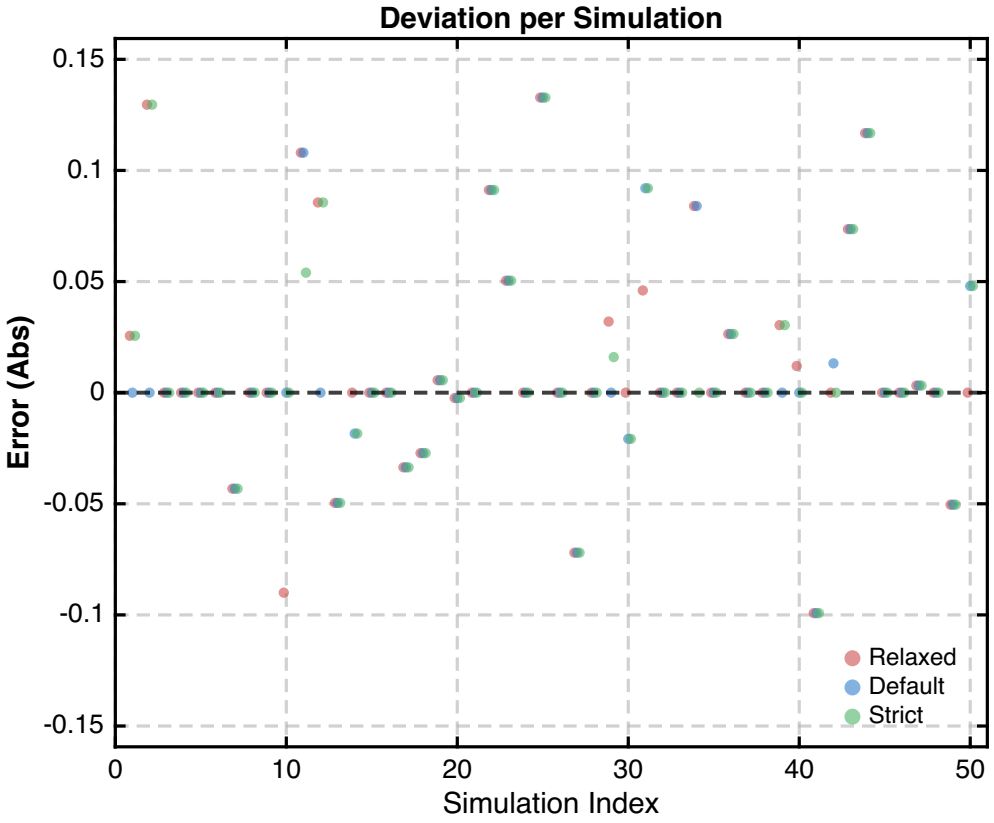


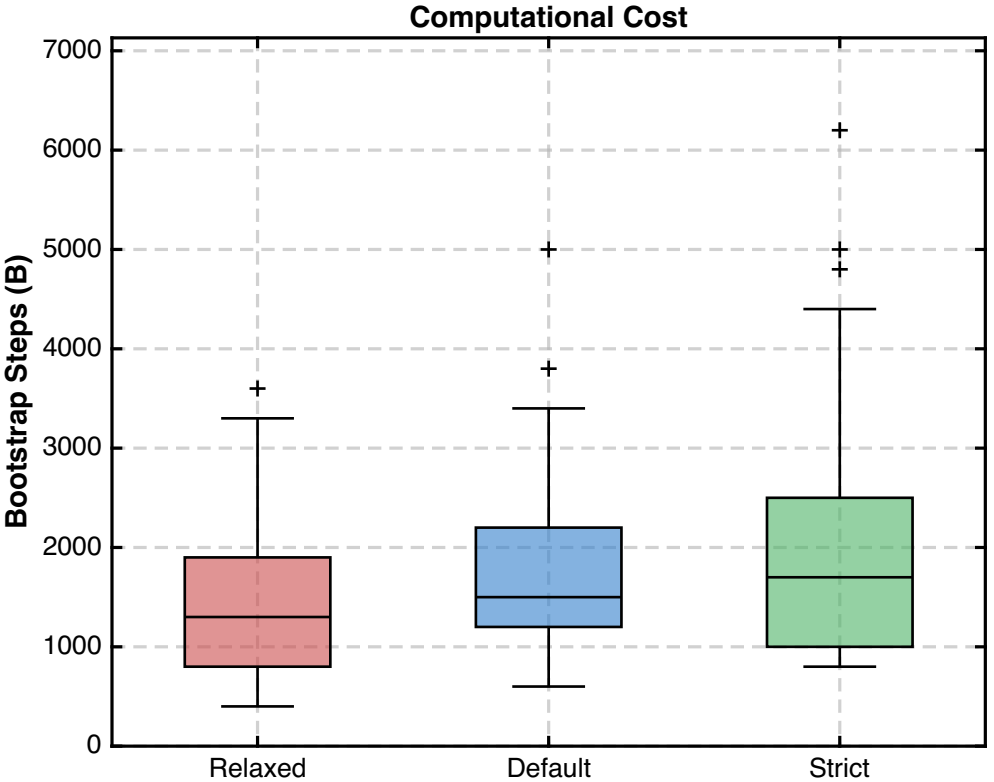
Ranking (Mean) Analysis: n 50 Bimodal (Median Cliff's d = 0.56)

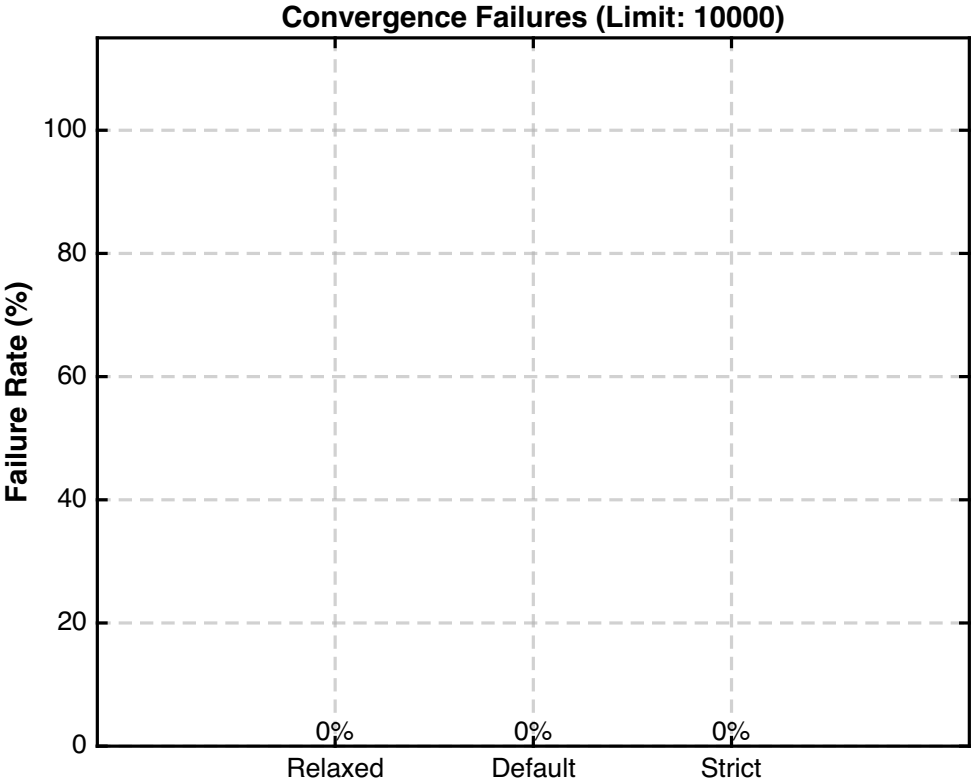


Thresholds (Delta) Analysis: n 50 Large Effect (Median Cliff's d = 0.86)

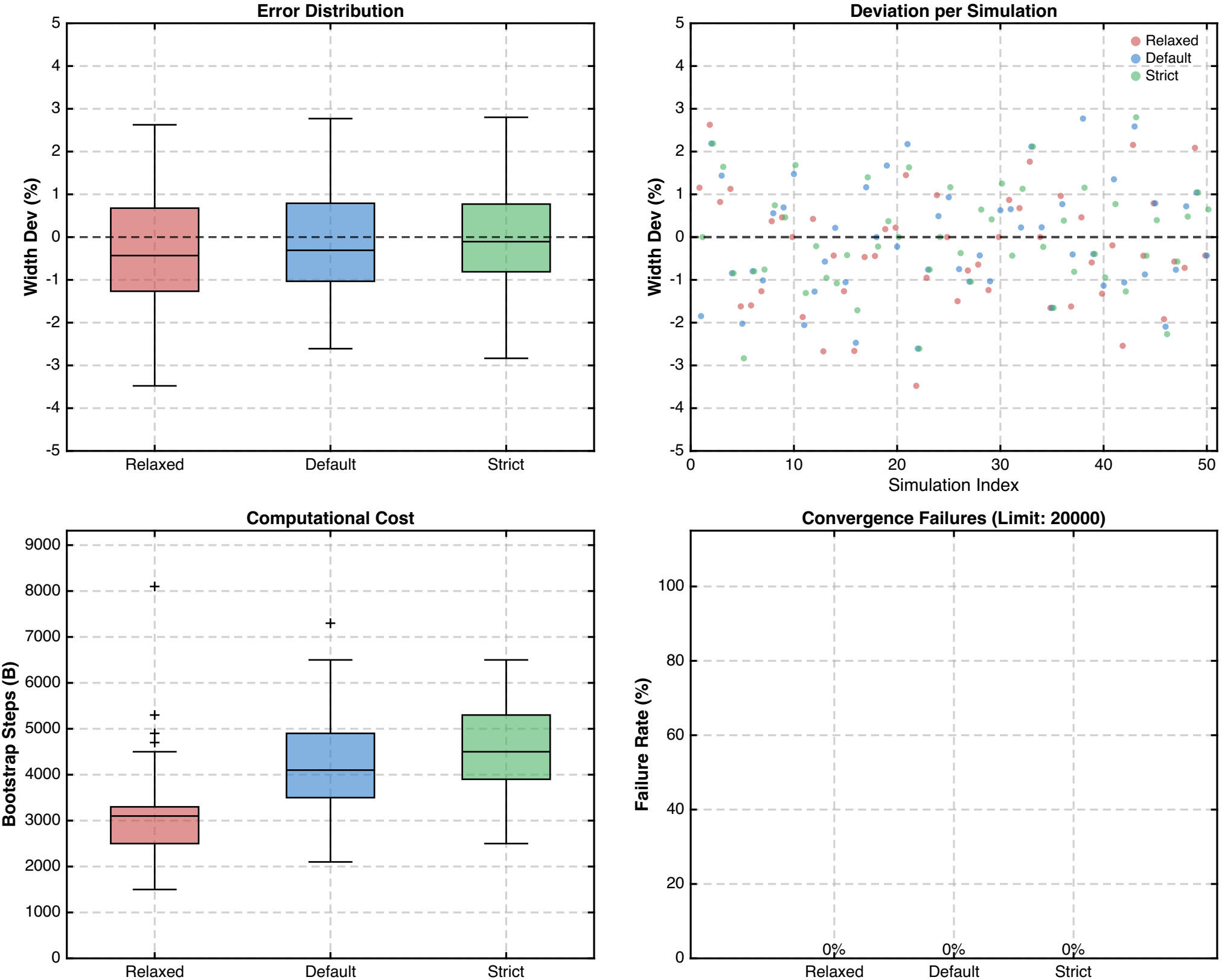
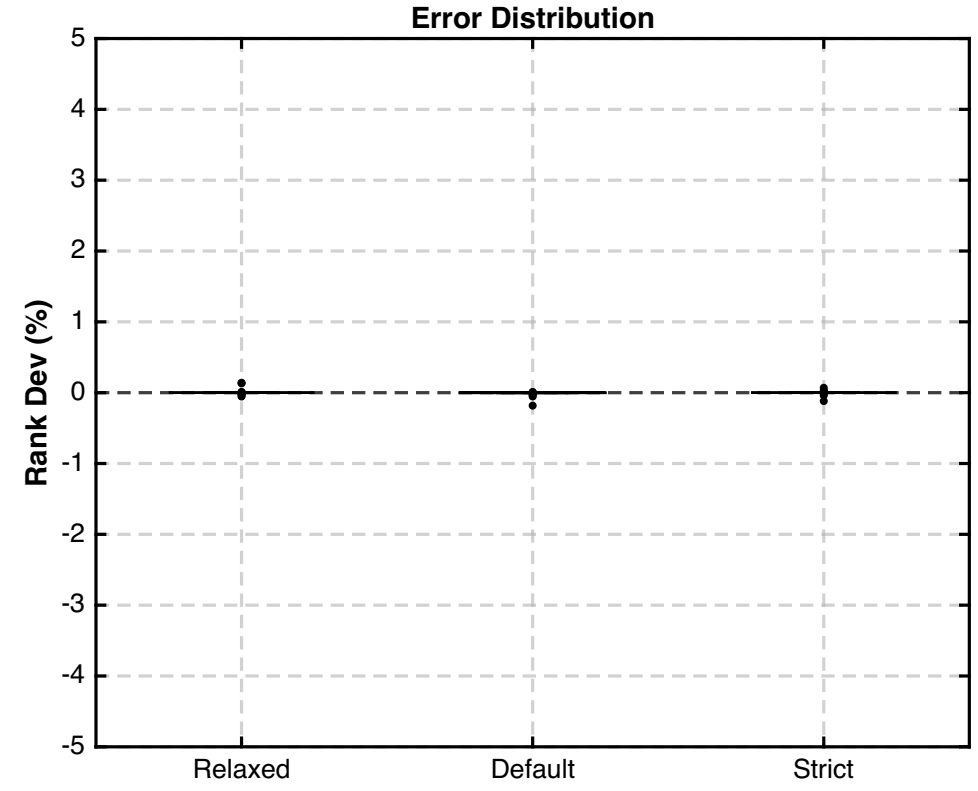
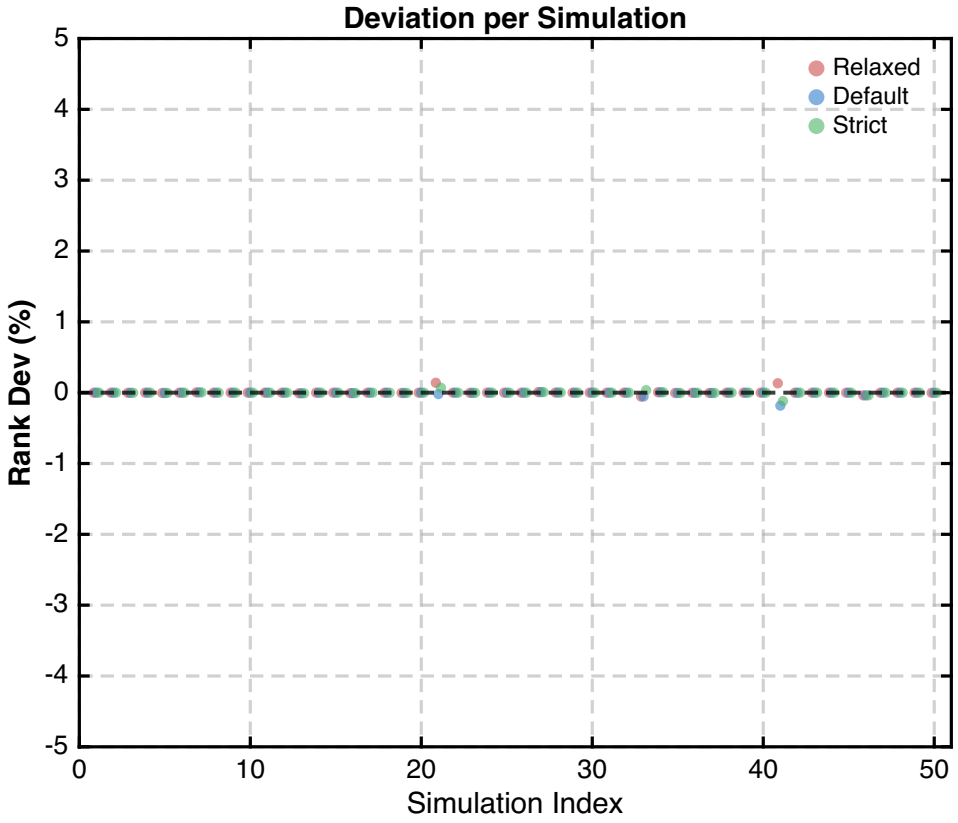
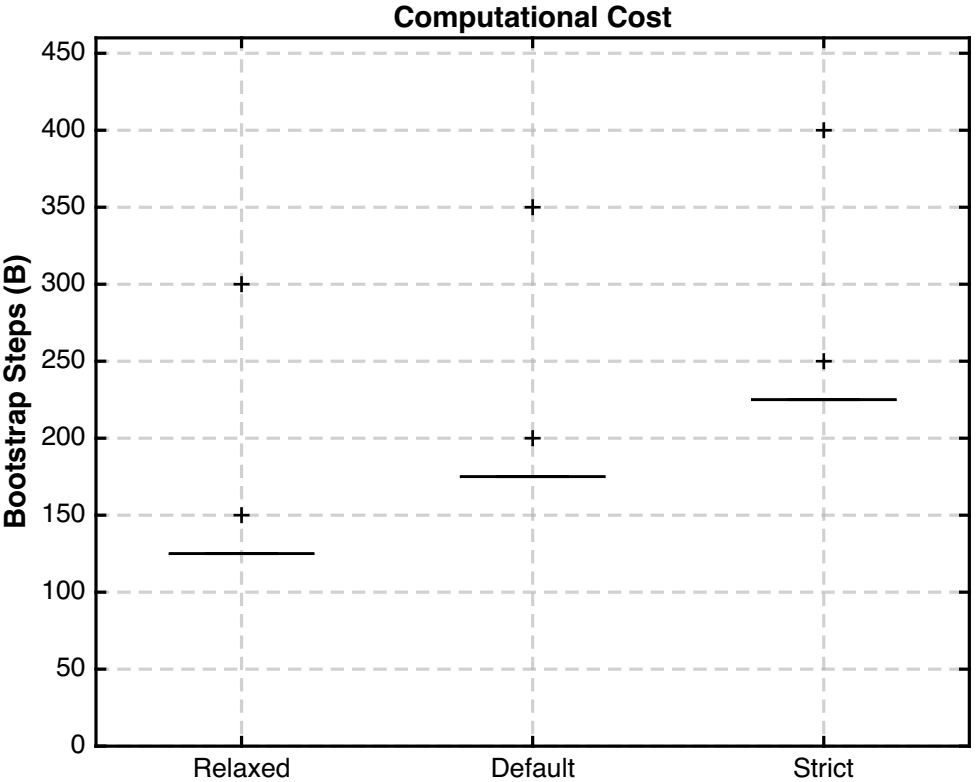
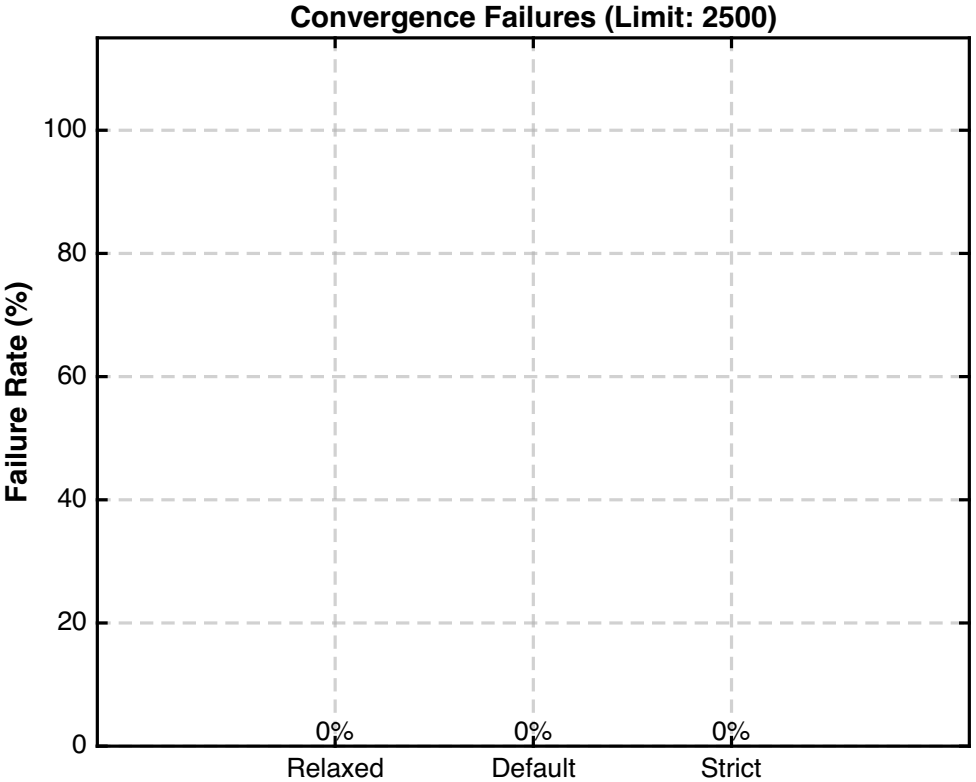








BCa CI (Width) Analysis: n 50 Large Effect (Median Cliff's d = 0.86)



Ranking (Mean) Analysis: n 50 Large Effect (Median Cliff's d = 0.86)